

Anomaly SUSY Transformation in SUSY Gauge Theory

The one-loop anomalous supersymmetry Ward identity of BPS letters in $\mathcal{N} = 4$ super-Yang–Mills: superspace, component, heat-kernel and holomorphic-twist approaches

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Abstract

We compute the one-loop anomaly of the supersymmetry transformation law — the anomalous supersymmetry Ward identity — acting on the semi-chiral (BPS-letter) composite operators of Euclidean $\mathcal{N} = 4$ super-Yang–Mills theory, quantized in $\mathcal{N} = 1$ superspace in a fixed Fermi–Feynman representative and regulated by dimensional reduction. The four letter families b , β_r , γ^r , $\partial_{\dot{\alpha}} c$ ($r = 1, 2, 3$) give nine component letters and eighty-one ordered composite insertions. The entire quantum correction is the failure of the regulated cutting rule, a single universal evanescent insertion $\mu_{\ell}^2/(D_0 D_1 D_2)$, and is therefore local. Twenty-nine ordered channels receive an exact nonzero local anomaly and fifty-two vanish exactly; every nonzero channel carries the universal coefficient $\hbar g^2/(16\pi^2)$ times the structure-constant tensor $f^{ACD} f^{BCE}$, and the full tower of holomorphic derivatives is generated in closed form by the kernel $K_{m,n;k,\ell} = 2 \binom{m}{k} \binom{n}{\ell} / [(m+n+2)(k+\ell+1)]$. We develop the result in four parallel frameworks. The superspace Feynman-diagram derivation is complete and constitutes the main result. The component Feynman-diagram approach yields one exact channel by bottom projection, $Q_{-}(\psi_{r+}^A \tilde{\phi}_s^B) = -(\sqrt{2} \hbar g^2/32\pi^2) \delta_{rs} f^{ACD} f^{BCE} \tilde{\chi}_{\dot{\alpha}}^D \tilde{\chi}^{E\dot{\alpha}}$, with its remaining sectors open. The heat-kernel approach supplies exact semigroup lemmas but is presently blocked: no single elliptic generator with the correct blockwise signs and normalizations has yet been constructed. Finally, the result agrees exactly — all eighty-one ordered output words and the derivative tower to all orders — with the holomorphic-twist prediction of Budzik–Kulp after correcting an internal factor-two inconsistency of that reference, the missing $\Gamma(3) = 2$ of the triangle parametrization; we discuss precisely what this agreement does and does not prove. All conventions — Lorentzian and Euclidean spinors, Weyl and Majorana fermions, covariant derivatives, θ -expansions, chiral and vector representations, and the BV–BRST quantization — are developed self-contained from the ground up.

Contents

1	Introduction	2
1.1	The anomalous supersymmetry transformation law	2
1.2	Main result	3
1.3	Four approaches, four statuses	4
1.4	Organization	5
2	Conventions: spinors, superspace and superfields	5
2.1	Lorentzian spinors	5
2.2	Euclidean spinors	6

2.3	Flat $N = 1$ superspace	7
2.4	Theta expansions and superfields	8
2.5	Chiral and vector representations	9
3	Supersymmetric gauge theory and BV–BRST quantization	10
3.1	Gauge and chiral superfields in the chiral representation	10
3.2	Chiral versus vector representation	11
3.3	Component reconstruction	11
3.4	$\mathcal{N} = 1$, $\mathcal{N} = 2$ and $\mathcal{N} = 4$ super-Yang–Mills	12
3.5	BV–BRST quantization	12
3.6	Gauge fixing and propagators	14
3.7	Regularization: dimensional reduction	16
4	The superspace Feynman-diagram derivation	16
4.1	The letters	16
4.2	The Schwinger–Dyson identity and the Euler-operator reduction	17
4.3	The cutting mechanism	18
4.4	Selection rules: the census of the eighty-one ordered pairs	20
4.5	The seed channel (b, b) in full	20
4.6	The complete channel table	22
4.7	The holomorphic-derivative tower	24
5	The component Feynman-diagram approach	25
5.1	The program: component quantization of the same action	25
5.2	The accepted result: bottom projection of the (β, γ) anomaly	26
5.3	Open items	27
6	The heat-kernel approach	28
6.1	Motivation: a semigroup-based second derivation	28
6.2	Established scalar and semigroup lemmas	28
6.3	The obstruction: the regulator proposal fails	30
6.4	What would close the route	30
7	Comparison with the holomorphic twist	31
7.1	The external target	31
7.2	The dictionary	32
7.3	Comparison results	33
7.4	Two source-internal inconsistencies and their adjudication	34
7.5	The admissibility verdict	35
8	Conclusions and open problems	36
A	Verification status	37

1 Introduction

1.1 The anomalous supersymmetry transformation law

Global supersymmetry Ward identities are among the most rigid constraints of quantum field theory: they organize composite operators into supermultiplets and underlie the non-renormalization properties of supersymmetric gauge theories [2]. A sharper question than the validity of such identities is whether the *transformation law itself* — the statement that a supercharge acts on

a composite operator through the classical variations of its factors — survives quantization, or whether it is deformed at loop level. When such a deformation is ultraviolet-finite, local in the external momenta, and scheme-independent, it is an anomaly of the supersymmetry transformation law, in the same lineage as the Konishi anomaly, rather than a renormalization effect.

This paper determines the complete one-loop answer for the maximally supersymmetric case: Euclidean $\mathcal{N} = 4$ super-Yang–Mills theory with a compact gauge algebra, quantized in $\mathcal{N} = 1$ superspace [3]. The sharp probe is supplied by the BPS letters and their semi-chiral composites. In the Euclidean plus/minus spin frame, the gauge-covariant superspace spinor derivative splits into two components $\nabla_{\pm}$, and the elementary fields assemble into nine BPS letters: one built from the chiral field-strength superfield, three from the chiral matter multiplets, three from the independent Euclidean antichiral multiplets, and two from the antichiral field strength (Secs. 2 and 4). At tree level the semi-chiral derivative ∇_- of every letter is either zero or a sum of equation-of-motion (Euler) operators and explicit composites, so the ordered products $L_i L_j$ — the semi-chiral composites — obey a closed Ward identity modulo equations of motion. Precisely because the tree reduction is exact and Euler insertions are eliminated by the Schwinger–Dyson identity, the one-loop correction to this transformation law is finite and local: any nonzero answer is an anomaly, not an artifact of subtraction. The same letters are the avatars of the 1/16-BPS sector under the holomorphic twist of the theory [7], where the one-loop deformation of the supercharge action on letter pairs has been computed from the holomorphic anomaly by Budzik and Kulp [1].

Microscopically, the entire anomaly originates in a single mechanism: dimensional reduction’s evanescent failure of the cutting rule [5]. In the superspace Schwinger–Dyson derivation, an Euler operator attached to a propagator collapses it, and every one-loop graph reduces to a cut line times a tree subgraph. In dimensional reduction (DRED) with $d = 4 - 2\epsilon$, the superspace spin algebra is kept four-dimensional while the propagator denominators are d -dimensional; the two structures are kept strictly separate and are never fused into one another before the Schwinger sum is performed. For every marked internal line of every one-loop graph, with four-dimensional numerator square $\bar{r}_e^2$ and d -dimensional denominators D_i , the cutting rule therefore fails by one universal evanescent insertion,

$$\frac{\bar{r}_e^2}{D_0 D_1 D_2} - \frac{1}{D_1 D_2} = \frac{\mu_\ell^2}{D_0 D_1 D_2}, \quad \lim_{\epsilon \rightarrow 0} \mu^{2\epsilon} \int \frac{d^{4-2\epsilon} \ell}{(2\pi)^{4-2\epsilon}} \frac{\mu_\ell^2}{(\ell^2 + \Delta)^3} = \frac{1}{32\pi^2}, \quad (1.1)$$

with no graph-specific factor of $4-d$. The evanescent master integral is finite because $\epsilon \Gamma(\epsilon) \rightarrow 1$, and its would-be nonlocal Δ -dependence enters only at order ϵ : the anomaly is a polynomial in the external momenta, i.e. the insertion of a local composite operator. Replacing the numerator by its d -dimensional form before the subtraction would annihilate (1.1) identically and erase the anomaly.

1.2 Main result

Consider Euclidean $\mathcal{N} = 4$ super-Yang–Mills theory, with gauge generators $[T_A, T_B] = i f_{AB}^C T_C$ and Killing form normalized to $\kappa_{AB} = \delta_{AB}$, quantized in $\mathcal{N} = 1$ superspace. The main result of this paper is the complete one-loop anomalous supersymmetry Ward identity of the BPS-letter composites: for each of the $9 \times 9 = 81$ ordered pairs (L_i, L_j) ,

$$\nabla_-(L_i^A L_j^B) \Big|_{1\text{-loop}} = \frac{\hbar g^2}{16\pi^2} f^{ACD} f^{BCE} \Delta(L_i, L_j)^{DE}, \quad (1.2)$$

with the sum over C understood. Three features are universal. First, the coefficient $\hbar g^2/(16\pi^2)$ and the ordered color tensor $f^{ACD} f^{BCE}$ are the same in every channel. Second, the letter bilinear $\Delta(L_i, L_j)^{DE}$ is fixed explicitly (Sec. 4): exactly 29 of the 81 ordered channels are

nonzero and the remaining 52 vanish exactly, for four independent structural reasons; reversed orderings (L_j, L_i) are independent rows, not quotients of (L_i, L_j) . Third, the anomaly is local, finite, and — up to a finite normal-product term fixed by the residual supersymmetries of the vacuum, under which the 29 nonzero channels close — scheme-independent.

Resumming the holomorphic-derivative tower, the anomaly of an arbitrary derivative composite is governed by the closed-form kernel

$$K_{m,n;k,\ell} = \frac{2 \binom{m}{k} \binom{n}{\ell}}{(m+n+2)(k+\ell+1)}, \quad 0 \leq k \leq m, \quad 0 \leq \ell \leq n, \quad (1.3)$$

where the overall factor 2 is $\Gamma(3)$ of the triangle Feynman parametrization, not an orientation multiplicity. The same 81 channels and the same tower were predicted, up to normalization, by the holomorphic-twist computation of [1]. After a factor-two correction of the source — its printed master-triangle and Appendix-B tower coefficients miss exactly $\Gamma(3) = 2$ relative to its own displayed component results — the agreement is exact on all 81 ordered output words and on the full derivative tower (Sec. 7). The comparison simultaneously adjudicates a second internal inconsistency of the source, whose compact one-loop coefficient is printed both as $-1/4$ and as $-\kappa^2$ with $\kappa^2 = 1/4$, selecting the main-text branch.

1.3 Four approaches, four statuses

The result (1.2) was attacked along four parallel routes, which have reached sharply different levels of completion; we state each status honestly, and Sec. 8 collects the open problems.

The first route is the direct *superspace Feynman-diagram* computation (Sec. 4): the Schwinger–Dyson reduction of every ordered letter pair, the occurrence-resolved cutting failure (1.1), and exact-arithmetic evaluation of all 81 channels.

Remark 1.1. The superspace route is complete and is the accepted derivation of (1.2): 81 channels computed exactly, 29 nonzero and 52 exactly zero, with no unresolved coefficient row. An adversarial census of potentially omitted graph sectors remains open as a completeness question; it does not touch any accepted coefficient (Sec. 7 and App. A).

The second route is the *component Feynman-diagram* formulation (Sec. 5), which has locked the exact Euclidean $\mathcal{N} = 4$ component actions, the BV–BRST data [4], and one exact physical channel: the bottom projection of the matter-fermion/antichiral-scalar composite,

$$Q_- \left(\psi_{r+}^A \tilde{\phi}_s^B \right) \Big|_{1\text{-loop}} = - \frac{\sqrt{2} \hbar g^2}{32\pi^2} \delta_{rs} f^{ACD} f^{BCE} \tilde{\chi}_\alpha^D \tilde{\chi}^{E\dot{\alpha}}, \quad (1.4)$$

whose dotted-gaugino bilinear is color-symmetric, so every antisymmetric color word annihilates it.

Remark 1.2. The component route is partial. Equation (1.4) is an exact projection of the superfield result, not an independent component derivation: the would-be ordinary two-Yukawa parent graph does not exist because the mixed propagators vanish identically. The complete component Euler operators and the announced termwise off-shell Noether checks, the complete component Feynman rules, an independent all-pairs component sweep, and the Lorentzian vector integration cycle all remain open (Sec. 8).

The third route is a non-perturbative *heat-kernel* derivation (Sec. 6) in the spirit of [6]: regulate the one-loop fluctuation determinant by the semigroup of a single elliptic operator and read the anomaly off a short-time coefficient.

Remark 1.3. The heat-kernel route has produced exact, machine-verified auxiliary lemmas — the ordered-simplex moment $1/((p+1)(p+q+2))$, the second-order semigroup/Duhamel coefficient, the vanishing short-time limit at fixed nonzero shift, and consistency with the master

integral of (1.1) — but the full derivation is blocked: the proposed regulator’s matter blocks give $16\Box\mathcal{P}_+$ and its vector block gives $+\Box$, against the declared $-\Box$ generator, and applying different powers of the operator on different blocks is not one functional calculus. The accepted coefficient does not come from this route.

The fourth route is the *holomorphic-twist* comparison (Sec. 7): translate the external one-loop prediction of [1] into the present alphabet and compare, word by word and jet by jet.

Remark 1.4. The holomorphic-twist agreement is settled as an external comparison only: the independently sealed coefficient table matches the factor-two-corrected prediction of [1] on all 81 ordered output words and all checked tower coefficients. Equality of totals is a cross-check, not an internal completeness proof — omitted contributions could cancel in the sum — and the 17-candidate adversarial census remains open. The twisted source is also not one-loop exact: its own algebra requires higher-loop corrections.

1.4 Organization

Section 2 fixes notation and conventions. Section 3 presents the gauge fixing and BV–BRST quantization. Section 4 contains the superspace derivation of (1.2), the 81-channel census, and the kernel (1.3). Section 5 treats the component formulation and proves (1.4). Section 6 presents the heat-kernel auxiliary lemmas and the blocker. Section 7 gives the holomorphic-twist comparison and its admissibility verdict. Section 8 summarizes and lists the open problems. Appendix A records the verification status: the machine-locked exact-arithmetic checks and the independent review trail.

2 Conventions: spinors, superspace and superfields

We use two-component Weyl spinors throughout, in both Lorentzian and Euclidean signature; textbook references for the superspace conventions are [3, 2]. Lorentzian vector indices are $\mu, \nu, \rho = 0, 1, 2, 3$, Euclidean vector indices are $m, n, r = 1, 2, 3, 4$, undotted spinor indices are $\alpha, \beta, \gamma = 1, 2$, and dotted spinor indices are $\dot{\alpha}, \dot{\beta}, \dot{\gamma} = \dot{1}, \dot{2}$. The index alphabet already distinguishes the two signatures; where confusion is possible we add the label L (Lorentzian) or E (Euclidean) explicitly. All commutators are graded, $[A, B] := AB - (-1)^{|A||B|}BA$, with $|X| \in \{0, 1\}$ the Grassmann parity.

2.1 Lorentzian spinors

We use the mostly-plus metric and the orientation

$$\eta_{\mu\nu} = \text{diag}(-1, +1, +1, +1), \quad v_\mu = \eta_{\mu\nu}v^\nu, \quad \epsilon^{0123} = +1. \quad (2.1)$$

The two-component epsilon tensors and index raising and lowering are

$$\epsilon^{12} = \epsilon^{\dot{1}\dot{2}} = +1, \quad \epsilon_{12} = \epsilon_{\dot{1}\dot{2}} = -1, \quad \psi^\alpha = \epsilon^{\alpha\beta}\psi_\beta, \quad \bar{\psi}^{\dot{\alpha}} = \epsilon^{\dot{\alpha}\dot{\beta}}\bar{\psi}_{\dot{\beta}}, \quad (2.2)$$

with $\epsilon^{\alpha\beta}\epsilon_{\beta\gamma} = \delta^\alpha_\gamma$, $\epsilon_{\alpha\beta}\epsilon^{\beta\gamma} = \delta_\alpha^\gamma$, and the same relations with dotted indices. Spinor contractions follow the *asymmetric* summation convention: undotted indices are contracted from upper left to lower right, dotted indices from lower left to upper right,

$$\xi\chi := \xi^\alpha\chi_\alpha, \quad \bar{\xi}\bar{\chi} := \bar{\xi}_{\dot{\alpha}}\bar{\chi}^{\dot{\alpha}}, \quad \xi^\alpha\chi_\alpha = -\xi_\alpha\chi^\alpha, \quad \bar{\xi}^{\dot{\alpha}}\bar{\chi}_{\dot{\alpha}} = -\bar{\xi}_{\dot{\alpha}}\bar{\chi}^{\dot{\alpha}}. \quad (2.3)$$

For Grassmann-odd spinors this gives $\xi\chi = \chi\xi$ and $\bar{\xi}\bar{\chi} = \bar{\chi}\bar{\xi}$. With $\sigma^1, \sigma^2, \sigma^3$ the standard Pauli matrices, the sigma matrices are

$$(\sigma^\mu)_{\alpha\dot{\alpha}} = (\mathbf{1}_2, \sigma^1, \sigma^2, \sigma^3)_{\alpha\dot{\alpha}}, \quad (\bar{\sigma}^\mu)^{\dot{\alpha}\alpha} := \epsilon^{\alpha\beta}\epsilon^{\dot{\alpha}\dot{\beta}}(\sigma^\mu)_{\beta\dot{\beta}} = (\mathbf{1}_2, -\sigma^1, -\sigma^2, -\sigma^3)^{\dot{\alpha}\alpha}, \quad (2.4)$$

and obey the Clifford relations

$$(\sigma^\mu)_{\alpha\dot{\alpha}}(\bar{\sigma}^\nu)^{\dot{\alpha}\beta} + (\sigma^\nu)_{\alpha\dot{\alpha}}(\bar{\sigma}^\mu)^{\dot{\alpha}\beta} = -2\eta^{\mu\nu}\delta_\alpha^\beta, \quad (\bar{\sigma}^\mu)^{\dot{\alpha}\alpha}(\sigma^\nu)_{\alpha\dot{\beta}} + (\bar{\sigma}^\nu)^{\dot{\alpha}\alpha}(\sigma^\mu)_{\alpha\dot{\beta}} = -2\eta^{\mu\nu}\delta_{\dot{\beta}}^{\dot{\alpha}}. \quad (2.5)$$

The (anti-)self-dual Lorentz generators are

$$(\sigma^{\mu\nu})_\alpha^\beta := \frac{1}{4}[(\sigma^\mu)_{\alpha\dot{\gamma}}(\bar{\sigma}^\nu)^{\dot{\gamma}\beta} - (\sigma^\nu)_{\alpha\dot{\gamma}}(\bar{\sigma}^\mu)^{\dot{\gamma}\beta}], \quad (\bar{\sigma}^{\mu\nu})^{\dot{\alpha}}_{\dot{\beta}} := \frac{1}{4}[(\bar{\sigma}^\mu)^{\dot{\alpha}\gamma}(\sigma^\nu)_{\gamma\dot{\beta}} - (\bar{\sigma}^\nu)^{\dot{\alpha}\gamma}(\sigma^\mu)_{\gamma\dot{\beta}}], \quad (2.6)$$

with components $\sigma^{0i} = -\frac{1}{2}\sigma^i$, $\bar{\sigma}^{0i} = +\frac{1}{2}\sigma^i$ and $\sigma^{ij} = \bar{\sigma}^{ij} = -\frac{i}{2}\epsilon^{ijk}\sigma^k$.

Weyl versus Majorana spinors. The formalism is strictly two-component. In Lorentzian signature the barred spinor is the Hermitian conjugate of the unbarred one,

$$\bar{\chi}^{\dot{\alpha}} := (\chi^\alpha)^\dagger; \quad (2.7)$$

this is the Majorana reality condition: a Majorana four-spinor is the pair $(\chi_\alpha, \bar{\chi}^{\dot{\alpha}})$, while a Weyl spinor is half of this data and is the fundamental object here. In particular the Lorentzian supercharges obey $(Q_\alpha)^\dagger = \bar{Q}_{\dot{\alpha}}$.

2.2 Euclidean spinors

The Wick rotation keeps the spatial coordinates and the Grassmann coordinates fixed and rotates the time coordinate; the generators acquire phases,

$$x_E^i = x_L^i, \quad x_E^4 = ix_L^0, \quad P_i^E = -iP_i^L, \quad P_4^E = P_L^0 = H, \quad Q_\alpha^E = -iQ_\alpha^L, \quad \bar{Q}_{\dot{\alpha}}^E = -i\bar{Q}_{\dot{\alpha}}^L. \quad (2.8)$$

The Euclidean sigma matrices and Clifford relations are

$$(\sigma^m)_{\alpha\dot{\alpha}} = (-i\sigma^1, -i\sigma^2, -i\sigma^3, \mathbf{1}_2)_{\alpha\dot{\alpha}}, \quad (\bar{\sigma}^m)^{\dot{\alpha}\alpha} := \epsilon^{\alpha\beta}\epsilon^{\dot{\alpha}\dot{\beta}}(\sigma^m)_{\beta\dot{\beta}} = (+i\sigma^1, +i\sigma^2, +i\sigma^3, \mathbf{1}_2)^{\dot{\alpha}\alpha}, \quad (2.9)$$

$$(\sigma^m)_{\alpha\dot{\alpha}}(\bar{\sigma}^n)^{\dot{\alpha}\beta} + (\sigma^n)_{\alpha\dot{\alpha}}(\bar{\sigma}^m)^{\dot{\alpha}\beta} = 2\delta^{mn}\delta_\alpha^\beta, \quad (2.10)$$

so that $(\bar{\sigma}^m)^{\dot{\alpha}\alpha} = ((\sigma^m)_{\alpha\dot{\alpha}})^\dagger$ as matrices. The (anti-)self-dual rotation generators $\sigma^{mn}, \bar{\sigma}^{mn}$ are defined as in (2.6) with $\mu \rightarrow m$, and have components

$$\sigma^{ij} = \bar{\sigma}^{ij} = \frac{i}{2}\epsilon^{ijk}\sigma^k, \quad \sigma^{4i} = +\frac{i}{2}\sigma^i, \quad \bar{\sigma}^{4i} = -\frac{i}{2}\sigma^i. \quad (2.11)$$

The $SO(4) \simeq SU(2) \times SU(2)$ structure. In Euclidean signature the undotted index α and the dotted index $\dot{\alpha}$ are fundamental indices of two *independent* $SU(2)$ factors: $(\sigma^{mn})_\alpha^\beta$ acts only on undotted spinors and $(\bar{\sigma}^{mn})^{\dot{\alpha}}_{\dot{\beta}}$ only on dotted ones. From (2.11), σ^{mn} and $\bar{\sigma}^{mn}$ are the (anti-)self-dual generators of $\mathfrak{so}(4) = \mathfrak{su}(2)_+ \oplus \mathfrak{su}(2)_-$, interchanged by $\sigma \leftrightarrow \bar{\sigma}$; the vector $(\sigma^m)_{\alpha\dot{\alpha}}$ is the $(\mathbf{2}, \mathbf{2})$.

No Euclidean conjugation. In Euclidean signature no dagger relation is imposed between Q_α and $\bar{Q}_{\dot{\alpha}}$: the two Weyl sectors are independent. Consequently the chiral and antichiral fields are independent Euclidean fields, not conjugates. This holds for the matter multiplets $(\Phi, \tilde{\Phi})$ and for the two gaugini $(\lambda_\alpha, \tilde{\lambda}_{\dot{\alpha}})$ alike, and it is used throughout this paper: the superspace calculation of Section 4 and the component reduction of Section 5 treat untilded and tilded fields as independent integration variables.

Remark 2.1. Standing convention: *no Euclidean conjugation relates Φ to $\tilde{\Phi}$* (nor λ to $\tilde{\lambda}$). The Lorentzian identifications $\tilde{\Phi} = \bar{\Phi}$, $\tilde{\lambda} = \bar{\lambda}$ are Wick-transported relations only; no intrinsic Euclidean reality condition is imposed anywhere in this paper.

The plus/minus frame. In the Euclidean computation of Sections 4–5 the undotted $SU(2)$ index is resolved in the frame $\alpha = (+, -)$: an undotted spinor X_α has components $X_\pm$, and correspondingly the gauge-covariant superspace derivatives ∇_α (undotted) and $\tilde{\nabla}_{\dot{\alpha}}$ (dotted) have frame components

$$\nabla_\pm := \nabla_{\alpha=\pm}, \quad \tilde{\nabla}_\pm := \tilde{\nabla}_{\dot{\alpha}=\pm}. \quad (2.12)$$

Here ∇_α carries the chiral gauge connection and $\tilde{\nabla}_{\dot{\alpha}}$ the antichiral one, adapted to the independent Euclidean antichiral fields. In the one-loop calculation every remaining spinor index is written dotted, the undotted $SU(2)$ index being carried by the symbol $+$: the vector appears through the bispinor map $(\sigma^m)_{+\dot{\alpha}}$ and $(\sigma^{mn})_{++}$ generates the undotted (plus) $SU(2)$.

Remark 2.2. Only the frame and the names $\nabla_\pm, \tilde{\nabla}_\pm$ are fixed here. The connection-level definitions of the gauge-covariant derivatives are deferred to Section 3, their use in the letter algebra to Section 4.

2.3 Flat $N = 1$ superspace

Flat $N = 1$ superspace has coordinates $z = (x^m, \theta^\alpha, \bar{\theta}_{\dot{\alpha}})$ (or $(x^\mu, \theta, \bar{\theta})$ in Lorentzian signature); the odd coordinates are distinct from the supersymmetry transformation parameters $\varepsilon, \bar{\varepsilon}$. We write

$$\theta \sigma^m \bar{\theta} := \theta^\alpha (\sigma^m)_{\alpha\dot{\beta}} \bar{\theta}^{\dot{\beta}}, \quad \theta \sigma^\mu \bar{\theta} := \theta^\alpha (\sigma^\mu)_{\alpha\dot{\beta}} \bar{\theta}^{\dot{\beta}}. \quad (2.13)$$

Odd derivatives are *left* derivatives, $\partial_\alpha := \vec{\partial}/\partial\theta^\alpha$, $\bar{\partial}^{\dot{\alpha}} := \vec{\partial}/\partial\bar{\theta}_{\dot{\alpha}}$, $\bar{\partial}_{\dot{\alpha}} := \epsilon_{\dot{\alpha}\dot{\beta}} \bar{\partial}^{\dot{\beta}}$, obeying the graded Leibniz rule and

$$\{\partial_\alpha, \theta^\beta\} = \delta_\alpha^\beta, \quad \{\bar{\partial}^{\dot{\alpha}}, \bar{\theta}_{\dot{\beta}}\} = \delta^{\dot{\alpha}}_{\dot{\beta}}, \quad \{\bar{\partial}_{\dot{\alpha}}, \bar{\theta}^{\dot{\beta}}\} = -\delta_{\dot{\alpha}}^{\dot{\beta}}. \quad (2.14)$$

Remark 2.3. The minus sign in the last anticommutator of (2.14) is a standard source of sign errors; it traces back to $\bar{\partial}_{\dot{\alpha}} = \epsilon_{\dot{\alpha}\dot{\beta}} \bar{\partial}^{\dot{\beta}}$ with $\epsilon_{12} = -1$.

A supersymmetry transformation acts on the coordinates as

$$\theta'^\alpha = \theta^\alpha + \varepsilon^\alpha, \quad \bar{\theta}'_{\dot{\alpha}} = \bar{\theta}_{\dot{\alpha}} + \bar{\varepsilon}_{\dot{\alpha}}, \quad x'^\mu = x^\mu + i\varepsilon^\alpha (\sigma^\mu)_{\alpha\dot{\beta}} \bar{\theta}^{\dot{\beta}} - i\theta^\alpha (\sigma^\mu)_{\alpha\dot{\beta}} \bar{\varepsilon}^{\dot{\beta}}. \quad (2.15)$$

We use the same symbols for the Noether charges and for their realizations as differential operators on superspace, the difference being clear from context. With $P_\mu = -i\partial_\mu$ the Lorentzian supercharges and supersymmetric covariant derivatives are

$$Q_\alpha = -i\partial_\alpha + (\sigma^\mu)_{\alpha\dot{\beta}} \bar{\theta}^{\dot{\beta}} \partial_\mu, \quad \bar{Q}_{\dot{\alpha}} = -i\bar{\partial}_{\dot{\alpha}} - \theta^\beta (\sigma^\mu)_{\beta\dot{\alpha}} \partial_\mu, \quad (2.16)$$

$$\mathcal{D}_\alpha = \partial_\alpha - i(\sigma^\mu)_{\alpha\dot{\beta}} \bar{\theta}^{\dot{\beta}} \partial_\mu, \quad \bar{\mathcal{D}}_{\dot{\alpha}} = \bar{\partial}_{\dot{\alpha}} + i\theta^\beta (\sigma^\mu)_{\beta\dot{\alpha}} \partial_\mu, \quad (2.17)$$

where $\mathcal{D}_\alpha, \bar{\mathcal{D}}_{\dot{\alpha}}$ anticommute with all supercharges. Their algebras are

$$\{Q_\alpha, \bar{Q}_{\dot{\beta}}\} = 2i(\sigma^\mu)_{\alpha\dot{\beta}} \partial_\mu = -2(\sigma^\mu)_{\alpha\dot{\beta}} P_\mu, \quad \{\mathcal{D}_\alpha, \bar{\mathcal{D}}_{\dot{\beta}}\} = 2i(\sigma^\mu)_{\alpha\dot{\beta}} \partial_\mu = -2(\sigma^\mu)_{\alpha\dot{\beta}} P_\mu, \quad (2.18)$$

with vanishing same-chirality anticommutators. The same relations hold for the Hilbert-space Noether charges; the normalization is fixed by Hamiltonian positivity, $\{Q_1, (Q_1)^\dagger\} + \{Q_2, (Q_2)^\dagger\} = 4H$, and the central charge vanishes. The Lorentzian chiral coordinates are

$$y^\mu := x^\mu - i\theta \sigma^\mu \bar{\theta}, \quad \tilde{y}^\mu := x^\mu + i\theta \sigma^\mu \bar{\theta} (= \bar{y}^\mu), \quad \bar{\mathcal{D}}_{\dot{\alpha}} y^\mu = 0, \quad \mathcal{D}_\alpha \tilde{y}^\mu = 0. \quad (2.19)$$

After Wick rotation the coordinate momentum is $P_m = -\partial_m$ and

$$Q_\alpha = -\partial_\alpha + (\sigma^m)_{\alpha\dot{\beta}} \bar{\theta}^{\dot{\beta}} \partial_m, \quad \bar{Q}_{\dot{\alpha}} = -\bar{\partial}_{\dot{\alpha}} - \theta^\beta (\sigma^m)_{\beta\dot{\alpha}} \partial_m, \quad (2.20)$$

$$\mathcal{D}_\alpha = \partial_\alpha + (\sigma^m)_{\alpha\dot{\beta}} \bar{\theta}^{\dot{\beta}} \partial_m, \quad \bar{\mathcal{D}}_{\dot{\alpha}} = \bar{\partial}_{\dot{\alpha}} - \theta^\beta (\sigma^m)_{\beta\dot{\alpha}} \partial_m,$$

with algebras

$$\{Q_\alpha, \bar{Q}_{\dot{\beta}}\} = +2(\sigma^m)_{\alpha\dot{\beta}}\partial_m = -2(\sigma^m)_{\alpha\dot{\beta}}P_m, \quad \{\mathcal{D}_\alpha, \bar{\mathcal{D}}_{\dot{\beta}}\} = -2(\sigma^m)_{\alpha\dot{\beta}}\partial_m = +2(\sigma^m)_{\alpha\dot{\beta}}P_m. \quad (2.21)$$

Same-chirality anticommutators vanish in both signatures, all mixed Q - $\mathcal{D}$ anticommutators vanish, and everything commutes with the momentum. The Euclidean chiral coordinates are

$$y^m := x^m + \theta\sigma^m\bar{\theta}, \quad \tilde{y}^m := x^m - \theta\sigma^m\bar{\theta}, \quad \bar{\mathcal{D}}_{\dot{\alpha}}y^m = 0, \quad \mathcal{D}_\alpha\tilde{y}^m = 0. \quad (2.22)$$

Remark 2.4 (Sign trap). In Lorentzian signature the Q - and $\mathcal{D}$ -algebras coincide, $\{Q, \bar{Q}\} = \{\mathcal{D}, \bar{\mathcal{D}}\} = -2\sigma^\mu P_\mu$; in Euclidean signature they differ by a relative minus sign, $\{Q, \bar{Q}\} = -2\sigma^m P_m$ versus $\{\mathcal{D}, \bar{\mathcal{D}}\} = +2\sigma^m P_m$. All sign conventions of the loop computation in Sections 4–5 are tied to (2.21).

2.4 Theta expansions and superfields

The basic monomials and their normalizations are

$$\theta^2 := \theta^\alpha\theta_\alpha = -2\theta^1\theta^2, \quad \bar{\theta}^2 := \bar{\theta}_{\dot{\alpha}}\bar{\theta}^{\dot{\alpha}} = 2\bar{\theta}_{\dot{1}}\bar{\theta}_{\dot{2}}, \quad \theta^\alpha\theta^\beta = -\frac{1}{2}\epsilon^{\alpha\beta}\theta^2, \quad \bar{\theta}_{\dot{\alpha}}\bar{\theta}_{\dot{\beta}} = -\frac{1}{2}\epsilon_{\dot{\alpha}\dot{\beta}}\bar{\theta}^2. \quad (2.23)$$

With $\mathcal{D}^2 := \mathcal{D}^\alpha\mathcal{D}_\alpha$ and $\bar{\mathcal{D}}^2 := \bar{\mathcal{D}}_{\dot{\alpha}}\bar{\mathcal{D}}^{\dot{\alpha}}$ one has, in either signature,

$$\mathcal{D}^2\theta^2 = -4, \quad \bar{\mathcal{D}}^2\bar{\theta}^2 = -4. \quad (2.24)$$

Writing $X| := X|_{\theta=\bar{\theta}=0}$, the component projectors are

$$[X]_F := -\frac{1}{4}\mathcal{D}^2 X|, \quad [\tilde{X}]_{\tilde{F}} := -\frac{1}{4}\bar{\mathcal{D}}^2 \tilde{X}|, \quad [Y]_D := \frac{1}{16}\mathcal{D}^2\bar{\mathcal{D}}^2 Y| \quad (\bar{\mathcal{D}}^2 \text{ acts first}), \quad (2.25)$$

and the full superspace measure is $\int d^4x d^4\theta Y := \int d^4x [Y]_D$.

Remark 2.5. A general (unconstrained) superfield enters this paper only through the projectors (2.25); its full fourteen-component expansion is never used. The vector multiplet is always reduced to Wess–Zumino gauge, (2.33) below.

A chiral superfield Φ and an antichiral superfield $\tilde{\Phi}$ obey

$$\bar{\mathcal{D}}_{\dot{\alpha}}\Phi = 0, \quad \mathcal{D}_\alpha\tilde{\Phi} = 0, \quad (2.26)$$

and have components (ϕ, ψ_α, F) and $(\tilde{\phi}, \tilde{\psi}_{\dot{\alpha}}, \tilde{F})$ given by

$$\phi := \Phi|, \quad \psi_\alpha := \frac{1}{\sqrt{2}}\mathcal{D}_\alpha\Phi|, \quad F := -\frac{1}{4}\mathcal{D}^2\Phi|; \quad \tilde{\phi} := \tilde{\Phi}|, \quad \tilde{\psi}_{\dot{\alpha}} := \frac{1}{\sqrt{2}}\bar{\mathcal{D}}_{\dot{\alpha}}\tilde{\Phi}|, \quad \tilde{F} := -\frac{1}{4}\bar{\mathcal{D}}^2\tilde{\Phi}|. \quad (2.27)$$

In Lorentzian signature $\tilde{\Phi} = \bar{\Phi}$ and $(\tilde{\phi}, \tilde{\psi}, \tilde{F}) = (\bar{\phi}, \bar{\psi}, \bar{F})$; in Euclidean signature $\tilde{\Phi}$ is independent of Φ (Remark 2.1). Their chiral-coordinate expansions are

$$\Phi(y, \theta) = \phi(y) + \sqrt{2}\theta^\alpha\psi_\alpha(y) + \theta^2 F(y), \quad \tilde{\Phi}(\tilde{y}, \bar{\theta}) = \tilde{\phi}(\tilde{y}) + \sqrt{2}\bar{\theta}_{\dot{\alpha}}\tilde{\psi}^{\dot{\alpha}}(\tilde{y}) + \bar{\theta}^2 \tilde{F}(\tilde{y}), \quad (2.28)$$

with $y, \tilde{y}$ from (2.19) or (2.22). Expanding in x -space gives, in Euclidean signature,

$$\Phi = \phi + \sqrt{2}\theta\psi + \theta^2 F + \theta\sigma^m\bar{\theta}\partial_m\phi - \frac{1}{\sqrt{2}}\theta^2(\partial_m\psi^\alpha)(\sigma^m)_{\alpha\dot{\beta}}\bar{\theta}^{\dot{\beta}} + \frac{1}{4}\theta^2\bar{\theta}^2\partial^m\partial_m\phi, \quad (2.29)$$

$$\tilde{\Phi} = \tilde{\phi} + \sqrt{2}\bar{\theta}\tilde{\psi} + \bar{\theta}^2\tilde{F} - \theta\sigma^m\bar{\theta}\partial_m\tilde{\phi} + \frac{1}{\sqrt{2}}\theta^\alpha(\sigma^m)_{\alpha\dot{\beta}}(\partial_m\tilde{\psi}^{\dot{\beta}})\bar{\theta}^2 + \frac{1}{4}\theta^2\bar{\theta}^2\partial^m\partial_m\tilde{\phi}, \quad (2.30)$$

where $\partial^m\partial_m$ is the Euclidean Laplacian; the Lorentzian counterparts follow from (2.28) with (2.19).

The vector multiplet is described by the prepotential $\mathcal{V} = \mathcal{V}^A T_A$ with $[T_A, T_B] = if_{AB}{}^C T_C$ in an orthonormal basis, $\kappa_{AB} = \delta_{AB}$. Wess–Zumino gauge is the residual gauge choice

$$\mathcal{V}| = \theta\mathcal{V}| = \bar{\theta}\mathcal{V}| = 0, \quad (2.31)$$

in which the prepotential reduces to

$$\mathcal{V}_{\text{WZ}} = -2\theta\sigma^\mu\bar{\theta}A_\mu + 2i\theta^2\bar{\theta}_{\dot{\alpha}}\bar{\lambda}^{\dot{\alpha}} - 2i\bar{\theta}^2\theta^\alpha\lambda_\alpha + \theta^2\bar{\theta}^2D \quad (\text{Lorentzian}), \quad (2.32)$$

$$\mathcal{V}_{\text{WZ}} = -2i\theta\sigma^m\bar{\theta}A_m + 2i\theta^2\bar{\theta}_{\dot{\alpha}}\tilde{\lambda}^{\dot{\alpha}} - 2i\bar{\theta}^2\theta^\alpha\lambda_\alpha + \theta^2\bar{\theta}^2D \quad (\text{Euclidean}), \quad (2.33)$$

with component fields $(A_m, \lambda_\alpha, \tilde{\lambda}_{\dot{\alpha}}, D)$; note that the Euclidean prepotential contains the *independent* antichiral gaugino $\tilde{\lambda}_{\dot{\alpha}}$. The gauge field is the ordered-commutator projection

$$A_\mu = \frac{1}{8}(\bar{\sigma}_\mu)^{\dot{\beta}\alpha}[\mathcal{D}_\alpha, \bar{\mathcal{D}}_{\dot{\beta}}]_{\text{ord}}\mathcal{V} \quad (\text{Lorentzian}), \quad A_m = \frac{i}{8}(\bar{\sigma}_m)^{\dot{\beta}\alpha}[\mathcal{D}_\alpha, \bar{\mathcal{D}}_{\dot{\beta}}]_{\text{ord}}\mathcal{V} \quad (\text{Euclidean}), \quad (2.34)$$

where $[\cdot, \cdot]_{\text{ord}}$ keeps the spinor derivatives in the displayed order. The gaugino field-strength superfields are

$$\mathcal{W}_\alpha := -\frac{1}{8}\bar{\mathcal{D}}^2[e^{-\mathcal{V}}(\mathcal{D}_\alpha e^\mathcal{V})], \quad \widetilde{\mathcal{W}}_{\dot{\alpha}} := +\frac{1}{8}\mathcal{D}^2[e^\mathcal{V}(\bar{\mathcal{D}}_{\dot{\alpha}} e^{-\mathcal{V}})], \quad (2.35)$$

chiral and antichiral respectively, $\bar{\mathcal{D}}_{\dot{\alpha}}\mathcal{W}_\beta = 0$, $\mathcal{D}_\alpha\widetilde{\mathcal{W}}_{\dot{\beta}} = 0$, with component projections

$$\lambda_\alpha := i\mathcal{W}_\alpha|, \quad \tilde{\lambda}_{\dot{\alpha}} := -i\widetilde{\mathcal{W}}_{\dot{\alpha}}|, \quad D := -\frac{1}{2}\mathcal{D}^\alpha\mathcal{W}_\alpha| = +\frac{1}{2}\bar{\mathcal{D}}^{\dot{\alpha}}\widetilde{\mathcal{W}}_{\dot{\alpha}}|. \quad (2.36)$$

In Lorentzian signature $\widetilde{\mathcal{W}}_{\dot{\alpha}} = \bar{\mathcal{W}}_{\dot{\alpha}}$ and $\tilde{\lambda}_{\dot{\alpha}} = \bar{\lambda}_{\dot{\alpha}}$; in Euclidean signature $(\mathcal{W}_\alpha, \widetilde{\mathcal{W}}_{\dot{\alpha}})$ and $(\lambda_\alpha, \tilde{\lambda}_{\dot{\alpha}})$ are independent.

2.5 Chiral and vector representations

Gauge transformations are generated by a chiral parameter Λ and an independent antichiral parameter $\bar{\Lambda}$,

$$h := e^{i\Lambda}, \quad \bar{h} := e^{i\bar{\Lambda}}, \quad \bar{\mathcal{D}}_{\dot{\alpha}}\Lambda = 0, \quad \mathcal{D}_\alpha\bar{\Lambda} = 0. \quad (2.37)$$

In the *chiral representation* superfields are functions of the chiral coordinates (2.19)–(2.22), and

$$\Phi' = h\Phi, \quad \tilde{\Phi}' = \tilde{\Phi}\bar{h}^{-1}, \quad (e^\mathcal{V})' = \bar{h}e^\mathcal{V}h^{-1}. \quad (2.38)$$

Thus the bridge $e^\mathcal{V}$ carries the entire gauge field in this frame: the combination $\tilde{\Phi}e^\mathcal{V}\Phi$ is gauge invariant, the gauge connection appears only through $e^{-\mathcal{V}}(\mathcal{D}_\alpha e^\mathcal{V})$ in the undotted spinor derivative, and the field strengths (2.35) transform homogeneously,

$$\mathcal{W}'_\alpha = h\mathcal{W}_\alpha h^{-1}, \quad \widetilde{\mathcal{W}}'_{\dot{\alpha}} = \bar{h}\widetilde{\mathcal{W}}_{\dot{\alpha}}\bar{h}^{-1}. \quad (2.39)$$

In the *vector representation* one works instead on the full $(x, \theta, \bar{\theta})$ superspace: all superspace derivatives carry connections, the matter fields are covariantly chiral, and gauge transformations act by a single real group element, so that the complexified parameters $h, \bar{h}$ and the explicit bridge disappear from the transformation laws. The two representations are related by a factorization of the bridge $e^\mathcal{V}$ and are physically equivalent; we use the chiral representation in what follows.

Remark 2.6. This subsection states the two frames at the superfield level only. The connection-level definitions of the gauge-covariant derivatives $\nabla_\alpha, \tilde{\nabla}_{\dot{\alpha}}$ of (2.12), the bridge factorization, the gauge action, and the BV–BRST formulation are deferred to Section 3.

3 Supersymmetric gauge theory and BV–BRST quantization

This section collects the classical and quantum input for the one-loop computation of Sections 4–5: the general $\mathcal{N} = 1$ gauge–chiral superfield action in the chiral representation (Section 3.1), its vector-frame reformulation (Section 3.2) and component form (Section 3.3), the $\mathcal{N} = 1, 2, 4$ super–Yang–Mills specializations (Section 3.4), and the Batalin–Vilkovisky (BV) quantization [4] with the Fermi–Feynman supergraph rules and the dimensional-reduction regulator (Sections 3.5–3.7). Two global conventions govern everything below. First, the gauge coupling is absorbed into the connection, $A_M = A_M^A T_A$, $\mathcal{D}_M = \partial_M - iA_M$; the canonical normalization is restored by $A_M = gA_M^{\text{can}}$ (similarly for $\lambda, \phi, \psi, \mathcal{D}, F$), so every non-topological density carries an overall factor g^{-2} : vertices are g -independent up to that common factor, while propagators carry g^2 . The compact gauge algebra is $[T_A, T_B] = if_{AB}{}^C T_C$ with orthonormal Killing metric $\kappa_{AB} = \delta_{AB}$, and $f^{ABC} := \delta^{CD} f_{AB}{}^D$ is totally antisymmetric. Second, in Euclidean signature every tilded field is independent of its untilded partner before a middle-dimensional integration contour is chosen; on the Lorentzian contour the tilde becomes the formal conjugate. The computation of this paper is Euclidean; Lorentzian ancestors are quoted where instructive.

3.1 Gauge and chiral superfields in the chiral representation

Chiral and antichiral matter superfields $\Phi^I, \tilde{\Phi}_I$, in a representation $\mathcal{R}_{\text{mat}}$ and its dual, obey $\bar{\mathcal{D}}_{\dot{\alpha}}\Phi^I = 0$, $\mathcal{D}_{\alpha}\tilde{\Phi}_I = 0$. Gauge transformations are generated by a chiral parameter $\Lambda = \Lambda^A T_A$ and an independent antichiral $\tilde{\Lambda}$ (on the Lorentzian contour $\tilde{\Lambda} = \bar{\Lambda}$):

$$h := e^{i\Lambda}, \quad \tilde{h} := e^{i\tilde{\Lambda}}, \quad \Phi' = h\Phi, \quad \tilde{\Phi}' = \tilde{\Phi}\tilde{h}^{-1}. \quad (3.1)$$

The real prepotential $\mathcal{V} = \mathcal{V}^A T_A$ enters only through the bridge $e^{\mathcal{V}}$, which dresses the flat spinor derivatives to gauge-covariant ones:

$$(e^{\mathcal{V}})' = \tilde{h}e^{\mathcal{V}}h^{-1}, \quad \Gamma_{\alpha} := e^{-\mathcal{V}}\mathcal{D}_{\alpha}e^{\mathcal{V}}, \quad \tilde{\Gamma}_{\dot{\alpha}} := e^{\mathcal{V}}\bar{\mathcal{D}}_{\dot{\alpha}}e^{-\mathcal{V}}, \quad \nabla_{\alpha} := \mathcal{D}_{\alpha} + \Gamma_{\alpha}, \quad \nabla'_{\alpha} = h\nabla_{\alpha}h^{-1}. \quad (3.2)$$

The gaugino field-strength superfields are

$$\mathcal{W}_{\alpha} := -\frac{1}{8}\bar{\mathcal{D}}^2(e^{-\mathcal{V}}\mathcal{D}_{\alpha}e^{\mathcal{V}}), \quad \widetilde{\mathcal{W}}_{\dot{\alpha}} := +\frac{1}{8}\mathcal{D}^2(e^{\mathcal{V}}\bar{\mathcal{D}}_{\dot{\alpha}}e^{-\mathcal{V}}), \quad (3.3)$$

chiral and antichiral respectively and covariant: $\bar{\mathcal{D}}_{\dot{\alpha}}\mathcal{W}_{\beta} = 0$, $\mathcal{D}_{\alpha}\widetilde{\mathcal{W}}_{\dot{\beta}} = 0$, $\mathcal{W}'_{\alpha} = h\mathcal{W}_{\alpha}h^{-1}$, $\widetilde{\mathcal{W}}'_{\dot{\alpha}} = \tilde{h}\widetilde{\mathcal{W}}_{\dot{\alpha}}\tilde{h}^{-1}$. Engineering dimensions are $[\Phi] = 1$, $[\mathcal{W}_{\alpha}] = 3/2$, $[\mathcal{V}] = 0$.

With the ordered Berezin projectors of Section 2, $[X]_F := -\frac{1}{4}\mathcal{D}^2 X$, $[\tilde{X}]_{\tilde{F}} := -\frac{1}{4}\bar{\mathcal{D}}^2 \tilde{X}$, $[Y]_D := \frac{1}{16}\mathcal{D}^2\bar{\mathcal{D}}^2 Y$ (bar derivatives acting first), the general local two-derivative action is determined by four independent data [2, 3]: a real gauged Kähler density $\mathcal{K}_g(\tilde{\Phi}, \Phi, e^{\mathcal{V}})$ of dimension 2, a chiral superpotential $W(\Phi)$ of dimension 3, a symmetric chiral kinetic matrix $f_{AB}(\Phi)$ of dimension 0, and a Fayet–Iliopoulos parameter ξ_A . The exact Euclidean action used throughout this paper is

$$S_E = - \int d^4x \left\{ [\mathcal{K}_g(\tilde{\Phi}, \Phi, e^{\mathcal{V}})]_D + [W(\Phi)]_F + [\widetilde{W}(\tilde{\Phi})]_{\tilde{F}} + \frac{1}{4}[f_{AB}(\Phi)\mathcal{W}^{A\alpha}\mathcal{W}_{\alpha}^B]_F + \frac{1}{4}[\tilde{f}_{AB}(\tilde{\Phi})\widetilde{\mathcal{W}}_{\dot{\alpha}}^A\widetilde{\mathcal{W}}^{B\dot{\alpha}}]_{\tilde{F}} + \xi_A[\mathcal{V}^A]_D \right\}, \quad (3.4)$$

the Wick image of its Lorentzian ancestor S_L , with $\mathcal{L}_E = -\mathcal{L}_L|_{\text{Wick}}$. Exact gauge invariance of (3.4) is equivalent to

$$X_A^I \partial_I W = 0, \quad X_C^I \partial_I f_{AB} - f_{CA}^D f_{DB} - f_{CB}^D f_{AD} = 0, \quad \xi_A f_{BC}^A = 0, \quad (3.5)$$

with the holomorphic Killing vectors $X_A^I := i(T_A)^I{}_J \Phi^J$. The canonical slice of (3.4) is

$$\mathcal{K}_g = \tilde{\Phi}_I (e^{\mathcal{V}})^I{}_J \Phi^J, \quad f_{AB} = \left(\frac{1}{g^2} - \frac{i\theta_{\text{YM}}}{8\pi^2} \right) \delta_{AB}, \quad \tilde{f}_{AB} = \left(\frac{1}{g^2} + \frac{i\theta_{\text{YM}}}{8\pi^2} \right) \delta_{AB}. \quad (3.6)$$

3.2 Chiral versus vector representation

The chiral representation is one of three gauge frames related by bridge factorization [3]. Two complexified bridges $\mathcal{B}, \tilde{\mathcal{B}}$ with

$$e^\mathcal{V} = \tilde{\mathcal{B}}\mathcal{B}, \quad \mathcal{B}' = k\mathcal{B}h^{-1}, \quad \tilde{\mathcal{B}}' = \tilde{h}\tilde{\mathcal{B}}k^{-1} \quad (3.7)$$

interpolate between the chiral frame, in which matter transforms by the complexified parameters $(h, \tilde{h})$, and the vector (real) frame, in which matter $\Phi^\mathcal{V} := \mathcal{B}\Phi$ transforms by the real group element k only, $\Phi^{\mathcal{V}'} = k\Phi^\mathcal{V}$, and chirality is covariant, $\tilde{\nabla}_\alpha^\mathcal{V}\Phi^\mathcal{V} = 0$. In the vector frame all superspace derivatives carry connections; imposing the conventional constraints

$$\mathcal{F}_{\alpha\beta}^\mathcal{V} = 0, \quad \mathcal{F}_{\dot{\alpha}\dot{\beta}}^\mathcal{V} = 0, \quad \mathcal{F}_{\alpha\dot{\beta}}^\mathcal{V} = 0 \quad (3.8)$$

solves them in terms of the bridges, $\nabla_\alpha^\mathcal{V} = \tilde{\mathcal{B}}^{-1} \circ \mathcal{D}_\alpha \circ \tilde{\mathcal{B}}$, $\tilde{\nabla}_{\dot{\alpha}}^\mathcal{V} = \mathcal{B} \circ \bar{\mathcal{D}}_{\dot{\alpha}} \circ \mathcal{B}^{-1}$, with the vector derivative defined by $\{\nabla_\alpha^\mathcal{V}, \tilde{\nabla}_{\dot{\beta}}^\mathcal{V}\} = -2(\sigma_E^m)_{\alpha\dot{\beta}}\mathcal{D}_m^\mathcal{V}$ in Euclidean signature. The two representations agree identically at the level of the action: the density identity

$$\tilde{\Phi}^\mathcal{V}\Phi^\mathcal{V} = \tilde{\Phi}e^\mathcal{V}\Phi \quad (3.9)$$

and its field-strength analog imply $S^\mathcal{V} = S^\mathcal{C}$ term by term, the value of (3.4). The symmetric-bridge gauge $\mathcal{B} = \tilde{\mathcal{B}} = e^{\mathcal{V}/2}$ makes the identification manifest.

Remark 3.1. The perturbative computation of this paper is formulated entirely in the chiral representation. In this frame the gauge orbit is parametrized by the chiral–antichiral pair $(\Lambda, \tilde{\Lambda})$, the Faddeev–Popov ghosts are chiral superfields (Section 3.5), the gauge functions $\mathcal{F}_\pm$ are chiral and antichiral, and every interaction vertex is an ordered word in the single real prepotential $\mathcal{V}$ (Section 3.6); by (3.9) no physics is lost. The vector frame enters only through the background-covariant derivatives used to define $\mathcal{F}_\pm$ and the background-field split.

3.3 Component reconstruction

The Wess–Zumino gauge is the slice in which $\mathcal{V}$ is at most quadratic; the normalized Euclidean representative is

$$\mathcal{V}^{\text{WZ}} = -2i\theta\sigma_E^m\bar{\theta}A_m + 2i\theta^2\bar{\theta}_{\dot{\alpha}}\tilde{\lambda}^{\dot{\alpha}} - 2i\bar{\theta}^2\theta^\alpha\lambda_\alpha + \theta^2\bar{\theta}^2\mathcal{D}, \quad (3.10)$$

with $\mathcal{D}$ the auxiliary field (not to be confused with the spinor derivatives $\mathcal{D}_\alpha, \bar{\mathcal{D}}_{\dot{\alpha}}$), $(\mathcal{V}^{\text{WZ}})^3 = 0$ so that $e^{\pm\mathcal{V}} = 1 \pm \mathcal{V} + \frac{1}{2}\mathcal{V}^2$, and ordinary gauge transformations as the residual group. Matter component fields are the ordered projections

$$\phi^I := \Phi^I|, \quad \psi_\alpha^I := \frac{1}{\sqrt{2}}\mathcal{D}_\alpha\Phi^I|, \quad F^I := -\frac{1}{4}\mathcal{D}^2\Phi^I|, \quad (3.11)$$

together with the tilded partners; the vector-multiplet components are

$$\lambda_\alpha := i\mathcal{W}_\alpha|, \quad \tilde{\lambda}_{\dot{\alpha}} := -i\tilde{\mathcal{W}}_{\dot{\alpha}}|, \quad \mathcal{D} := -\frac{1}{2}\mathcal{D}^\alpha\mathcal{W}_\alpha| = +\frac{1}{2}\bar{\mathcal{D}}_{\dot{\alpha}}\tilde{\mathcal{W}}^{\dot{\alpha}}|, \quad \mathcal{D}_{(\alpha}\mathcal{W}_{\beta)}| = -i(\sigma_E^{mn})_{\alpha\beta}F_{mn}, \quad (3.12)$$

the equality of the two $\mathcal{D}$ projections being the superspace Bianchi identity. In the canonical slice (3.6) the Euclidean component density $\mathcal{L}_E = \mathcal{L}_{E,\text{matter}} + \mathcal{L}_{E,W} + \mathcal{L}_{E,\text{gauge}}$ reads

$$\begin{aligned} \mathcal{L}_{E,\text{can}} = & (\mathcal{D}_m\tilde{\phi})^I(\mathcal{D}_m\phi)^I + \tilde{\psi}_{\dot{\alpha},I}(\bar{\sigma}_E^m)^{\dot{\alpha}\alpha}(\mathcal{D}_m\psi_\alpha)^I - \tilde{F}_IF^I - \mu_A^{(0)}\mathcal{D}^A \\ & - i\sqrt{2}[\tilde{\phi}_I(T_A)^I{}_J\lambda^{A\alpha}\psi_\alpha^J - \tilde{\psi}_{\dot{\alpha},I}(T_A)^I{}_J\tilde{\lambda}^{A\dot{\alpha}}\phi^J] \\ & - W_I(\phi)F^I + \frac{1}{2}W_{IJ}(\phi)\psi^{I\alpha}\psi_\alpha^J - \tilde{W}^{,IJ}(\tilde{\phi})\tilde{F}_I + \frac{1}{2}\tilde{W}^{,IJ}(\tilde{\phi})\tilde{\psi}_{\dot{\alpha},I}\tilde{\psi}_{\dot{J}}^{\dot{\alpha}} \\ & + \frac{1}{4g^2}F_{mn}^AF_{mn}^A + \frac{1}{g^2}\tilde{\lambda}_{\dot{\alpha}}^A(\bar{\sigma}_E^m)^{\dot{\alpha}\alpha}(\mathcal{D}_m\lambda_\alpha)^A - \frac{1}{2g^2}\mathcal{D}^A\mathcal{D}^A + \frac{i\theta_{\text{YM}}}{64\pi^2}\epsilon_E^{mnrs}F_{mn}^AF_{rs}^A, \end{aligned} \quad (3.13)$$

with $\mu_A^{(0)} := \tilde{\phi}_I(T_A)^I{}_J \phi^J + \xi_A$ and $\epsilon_E^{1234} = +1$. The auxiliary fields enter algebraically; their equations of motion

$$F^I = -\widetilde{W}^I(\tilde{\phi}), \quad \tilde{F}_I = -W_I(\phi), \quad \mathcal{D}^A = -g^2 \delta^{AB} \mu_B^{(0)} \quad (3.14)$$

complete the squares of the positive bosonic contour.

Remark 3.2. The complete off-shell component map — including the general Kähler geometry of the target, the moment map, and the shifted auxiliaries — has been reconstructed in the source repository and machine-verified coefficient by coefficient (336 independent component coefficients, zero failures); only the canonical slice is displayed here. The component-field route to the anomaly (Section 5) uses this map and is partial in the sense stated there.

3.4 $\mathcal{N} = 1$, $\mathcal{N} = 2$ and $\mathcal{N} = 4$ super-Yang–Mills

Pure $\mathcal{N} = 1$ super-Yang–Mills is the vector multiplet $(A_m^A, \lambda_\alpha^A, \tilde{\lambda}_{\dot{\alpha}}^A, \mathcal{D}^A)$ with action

$$S_E^{(1)} = -\frac{1}{4} \int d^4x ([f_{AB} \mathcal{W}^{A\alpha} \mathcal{W}_\alpha^B]_F + [\tilde{f}_{AB} \widetilde{\mathcal{W}}_{\dot{\alpha}}^A \widetilde{\mathcal{W}}^{B\dot{\alpha}}]_{\tilde{F}}). \quad (3.15)$$

$\mathcal{N} = 2$ super-Yang–Mills adds one adjoint chiral multiplet $\Phi = (\phi, \psi_\alpha, F)$; closure of the second supersymmetry at the linearized level, together with gauge invariance, fixes the relative coefficient of the matter kinetic term uniquely, giving

$$S_E^{(2)} = - \int d^4x \left\{ \frac{1}{g^2} \delta_{AB} [\tilde{\Phi}^A (e^{\mathcal{V}_{\text{ad}}})^B{}_C \Phi^C]_D + \frac{1}{4} [f_{AB} \mathcal{W}^{A\alpha} \mathcal{W}_\alpha^B]_F + \frac{1}{4} [\tilde{f}_{AB} \widetilde{\mathcal{W}}_{\dot{\alpha}}^A \widetilde{\mathcal{W}}^{B\dot{\alpha}}]_{\tilde{F}} \right\}, \quad (3.16)$$

with $e^{\mathcal{V}_{\text{ad}}}$ the bridge in the adjoint representation. $\mathcal{N} = 4$ super-Yang–Mills, the theory studied in this paper, contains three adjoint chirals Φ_r , $r = 1, 2, 3$, an $SU(3)$ flavor triplet inside $SU(4)_R$, together with the $SU(3)$ -invariant superpotential

$$W = -\frac{\sqrt{2}}{6g^2} \varepsilon_{rst} f^{ABC} \Phi_r^A \Phi_s^B \Phi_t^C, \quad (3.17)$$

whose coefficient $-\sqrt{2}/6$ is fixed uniquely by matching the gauge Yukawa coupling. The action is

$$S_E^{(4)} = - \int d^4x \left\{ \frac{1}{g^2} \delta_{AB} \sum_{r=1}^3 [\tilde{\Phi}_r^A (e^{\mathcal{V}_{\text{ad}}})^B{}_C \Phi_r^C]_D + \frac{1}{4} [f_{AB} \mathcal{W}^{A\alpha} \mathcal{W}_\alpha^B]_F + \frac{1}{4} [\tilde{f}_{AB} \widetilde{\mathcal{W}}_{\dot{\alpha}}^A \widetilde{\mathcal{W}}^{B\dot{\alpha}}]_{\tilde{F}} + [W]_F + [\widetilde{W}]_{\tilde{F}} \right\}. \quad (3.18)$$

The unified $\mathcal{N} = 1, 2, 4$ form, with multiplet counts $m_1 = 0$, $m_2 = 1$, $m_4 = 3$, is the starting point of the vertex inventory of Section 3.6.

3.5 BV–BRST quantization

Because the even noncompact gauge-orbit integral diverges and the odd one vanishes, gauge fixing is mandatory, and we use the BV formalism [4]. The minimal sector consists of the physical fields and a chiral–antichiral pair of Faddeev–Popov ghost superfields $c, \tilde{c}$; the non-minimal sector consists of an antighost–Nakanishi–Lautrup doublet $(\bar{c}_+, b_+)$ of chiral superfields and its antichiral partner $(\bar{c}_-, b_-)$, with $s\bar{c}_\pm = b_\pm$, $sb_\pm = 0$. Every field X carries a Grassmann parity ϵ_X , a ghost number and an engineering dimension; its BV antifield X^* (never complex conjugation) obeys

$$\epsilon_{X^*} = \epsilon_X + 1, \quad \text{gh}(X^*) = -1 - \text{gh}(X), \quad [X^*] = d_\Sigma - [X], \quad (3.19)$$

with $d_\Sigma = 2$ for the full superspace and $d_\Sigma = 3$ for the chiral and antichiral subspaces. The complete spectrum is

X	domain	ϵ_X	$\text{gh}(X)$	$[X]$
$\mathcal{V}$	$(x, \theta, \bar{\theta})$	0	0	0
Φ^I	(y, θ)	0	0	1
$\tilde{\Phi}_I$	$(\tilde{y}, \bar{\theta})$	0	0	1
c	(y, θ)	1	+1	0
$\tilde{c}$	$(\tilde{y}, \bar{\theta})$	1	+1	0
$\bar{c}_\pm$	$(y, \theta), (\tilde{y}, \bar{\theta})$	1	-1	2
$b_\pm$	$(y, \theta), (\tilde{y}, \bar{\theta})$	0	0	2

(3.20)

The classical BRST differential s ($\epsilon = 1$, $\text{gh} = 1$) acts as

$$se^\mathcal{V} = i\tilde{c}e^\mathcal{V} - ie^\mathcal{V}c, \quad sc = ic^2, \quad s\tilde{c} = i\tilde{c}^2, \quad s\Phi = ic\Phi, \quad s\tilde{\Phi} = -i\tilde{\Phi}\tilde{c}, \quad (3.21)$$

in components $sc^C = -\frac{1}{2}f_{AB}^C c^A c^B$, and is nilpotent on the untruncated local field algebra: $s^2 = 0$ and $sS_0 = 0$. The exact transformation of the prepotential is the Bernoulli series

$$s\mathcal{V} = \frac{\text{ad}_\mathcal{V}}{1 - e^{-\text{ad}_\mathcal{V}}} (ie^{-\text{ad}_\mathcal{V}}\tilde{c} - ic) = i \sum_{p,q \geq 0} \frac{B_{p+q}}{p!q!} [(-1)^q \mathcal{V}^p \tilde{c} \mathcal{V}^q - (-1)^p \mathcal{V}^p c \mathcal{V}^q], \quad (3.22)$$

with $B_0 = 1$, $B_1 = -\frac{1}{2}$, $B_2 = \frac{1}{6}$, $B_3 = 0$.

With the ordered pairings $\langle X, Y \rangle_8 := \int d^4x [X_A Y^A]_D$ and their chiral/antichiral analogs $\langle \cdot, \cdot \rangle_\pm$, the antibracket and the minimal master action are

$$(F, G) := \sum_X \int_{\Sigma_X} \left[F \frac{\overleftarrow{\delta}}{\delta X} \frac{\vec{\delta} G}{\delta X^*} - F \frac{\overleftarrow{\delta}}{\delta X^*} \frac{\vec{\delta} G}{\delta X} \right], \quad (3.23)$$

$$S_{\min} := S_0 + \langle \mathcal{V}^*, s\mathcal{V} \rangle_8 + \langle \Phi^*, s\Phi \rangle_+ + \langle \tilde{\Phi}^*, s\tilde{\Phi} \rangle_- - \langle c^*, sc \rangle_+ - \langle \tilde{c}^*, s\tilde{c} \rangle_-, \quad (3.24)$$

supplemented by the non-minimal stabilization $S_{\text{nm}} = S_{\min} + \langle \bar{c}^*, b \rangle$. They obey the classical master equation and generate s as a Hamiltonian vector field:

$$\frac{1}{2}(S_{\min}, S_{\min}) = 0, \quad sF = (S_{\min}, F), \quad s^2 F = \frac{1}{2}((S_{\min}, S_{\min}), F) = 0. \quad (3.25)$$

For the background split $e^{\mathcal{V}_{\text{tot}}} = \tilde{\mathcal{B}}_{\text{B}} e^{\mathcal{V}_{\text{q}}} \mathcal{B}_{\text{B}}$ the gauge functions are the background-covariant chiral and antichiral projections of the quantum prepotential,

$$\mathcal{F}_+ := -\frac{1}{4}(\bar{\nabla}_{\text{B}}^{\text{V}})^2 \mathcal{V}_{\text{q}}, \quad \bar{\nabla}_{\text{B}\dot{\alpha}}^{\text{V}} \mathcal{F}_+ = 0, \quad \mathcal{F}_- := -\frac{1}{4}(\nabla_{\text{B}}^{\text{V}})^2 \mathcal{V}_{\text{q}}, \quad \nabla_{\text{B}\alpha}^{\text{V}} \mathcal{F}_- = 0, \quad [\mathcal{F}_\pm] = 1. \quad (3.26)$$

With the collective pairing of the chiral/antichiral sectors, the standard gauge fermion and its BRST variation are

$$\Psi := \langle \bar{c}, \mathcal{F} \rangle - \frac{1}{2} \langle \bar{c}, \mathcal{Y}b \rangle, \quad s\Psi = \langle b, \mathcal{F} \rangle - \langle \bar{c}, \mathcal{M}^{\text{FP}} c^{\text{pair}} \rangle - \frac{1}{2} \langle b, \mathcal{Y}b \rangle + \frac{1}{2} \langle \bar{c}, (s\mathcal{Y})b \rangle, \quad (3.27)$$

where $\mathcal{Y}$ is an even, invertible, local, background-covariant kernel of dimension -1 , $c^{\text{pair}} := (c, \tilde{c})$ dressed to the background, and the Faddeev–Popov operator $\mathcal{M}^{\text{FP}}$ (even, ghost-number zero, dimension 1) is the gauge tangent of the quantum field, $s\mathcal{F} = \mathcal{M}^{\text{FP}} c^{\text{pair}}$. Integrating out the non-minimal doublets produces the Faddeev–Popov Berezinian

$$\mathfrak{F}^{\text{FP}} = \text{Ber}(\mathcal{M}_0^{-1} \mathcal{M}^{\text{FP}}), \quad (3.28)$$

evaluated on the quotient by the residual stabilizer, and, whenever the kernel $\mathcal{Y}$ is itself gauge averaged, a Nielsen–Kallosh factor fixed by the requirement that the multiplier Gaussian times

it reproduce the declared gauge-average normalization. At vanishing external antifield sources the gauge-fixed action is $S_\Psi = S_0 + s\Psi$.

The quantum master action $W = S_{\min} + \sum_{n \geq 1} \hbar^n M_n$ must satisfy the quantum master equation

$$\frac{1}{2}(W, W) - i\hbar\Delta W = 0 \quad (\text{Lorentzian}), \quad \frac{1}{2}(W, W) - \hbar\Delta W = 0 \quad (\text{Euclidean}), \quad (3.29)$$

with Δ the BV Laplacian of the measure; a nonzero left-hand side is the regulated BV anomaly functional. Expanded in $\hbar$ this is a recursion whose first (one-loop) rung in the Euclidean theory reads $sM_1 = \Delta S_{\min}$.

All Ward identities descend from the single Schwinger–Dyson identity of the regulated measure: for every coefficient mode q of the cylindrical expansion, with measure density ϖ and total path-integral exponent $\mathcal{E}_{\text{tot}}$,

$$0 = \int d\mu \, e^{\mathcal{E}_{\text{tot}}} \left[\varpi^{-1} \frac{\vec{\partial} \varpi}{\partial q} + \frac{\vec{\partial} \mathcal{E}_{\text{tot}}}{\partial q} \right]. \quad (3.30)$$

BRST, background and supersymmetry Ward identities are specializations of (3.30) to the corresponding vector fields. The anomalous supersymmetry Ward identity of this paper is the statement that the supersymmetry specialization of (3.30) retains a nonzero defect at one loop; it is constructed in Section 4.

Remark 3.3. At a finite mode cutoff the classical master equation is not automatic: the truncated master action obeys $\frac{1}{2}(S_{\min, \nu}, S_{\min, \nu})_\nu = s_\nu S_{0, \nu} + \sum (-1)^\epsilon Q^* s_\nu^2 Q$, whose right-hand side vanishes only after regulator closure, regulator invariance of the action, and nilpotency of the truncated differential are established. The identities above are therefore always understood with their regulated defect terms retained.

3.6 Gauge fixing and propagators

We now fix the perturbative scheme used in Sections 4–6. The Fermi–Feynman representative is the kernel choice

$$\mathcal{Y}_{\text{FF}}^{-1} = \frac{1}{2g^2} \mathcal{A}, \quad \mathcal{A} := \begin{pmatrix} 0 & -\bar{\mathcal{D}}^2/4 \\ -\mathcal{D}^2/4 & 0 \end{pmatrix}, \quad \mathcal{A}^2 = \square \mathbf{1}, \quad \mathcal{Y}_{\text{FF}} = 2g^2 \frac{\mathcal{A}}{\square}, \quad (3.31)$$

acting on the gauge-function doublet $(\mathcal{F}_+, \mathcal{F}_-)$; it produces the gauge-fixing term $\frac{1}{2} \langle \mathcal{Y}_{\text{FF}}^{-1} \mathcal{F}, \mathcal{F} \rangle$.

Remark 3.4. Two scheme choices are declared here. First, $\mathcal{Y}_{\text{FF}}$ contains $\square^{-1}$ and is therefore nonlocal as a gauge-fermion kernel: perturbation theory is *defined* by Fermi–Feynman Gaussian averaging rather than derived from a local admissible gauge fermion. (The source repository explicitly records this locality obstruction and does not assert a unique propagator set, a selected Nielsen–Kallosh branch, or momentum-space rules; the rules below are the standard Fermi–Feynman supergraph grammar [3], with coefficients that agree with the verified repo expressions.) Second, because $\mathcal{Y}_{\text{FF}}$ is field independent, $s\mathcal{Y}_{\text{FF}} = 0$ and the associated Nielsen–Kallosh factor is a field-independent constant in this slice; it cancels in normalized correlation functions and is set to unity.

With $\square := \delta^{mn} \partial_m \partial_n$ and $z = (x, \theta, \bar{\theta})$, the chiral, antichiral and transverse projectors are

$$\mathcal{P}_+ := \frac{\bar{\mathcal{D}}^2 \mathcal{D}^2}{16 \square}, \quad \mathcal{P}_- := \frac{\mathcal{D}^2 \bar{\mathcal{D}}^2}{16 \square}, \quad \mathcal{P}_T := -\frac{\mathcal{D}^\alpha \bar{\mathcal{D}}^2 \mathcal{D}_\alpha}{8 \square}, \quad \mathcal{P}_0 := \mathcal{P}_+ + \mathcal{P}_-, \quad (3.32)$$

with the projector algebra following from the spinor-derivative identities:

$$\mathcal{P}_i \mathcal{P}_j = \delta_{ij} \mathcal{P}_i \quad (i, j \in \{T, +, -\}), \quad \mathcal{P}_T + \mathcal{P}_+ + \mathcal{P}_- = 1, \quad (3.33)$$

$$\bar{\mathcal{D}}^2 \mathcal{D}^2 \bar{\mathcal{D}}^2 = 16 \square \bar{\mathcal{D}}^2, \quad \mathcal{D}^2 \bar{\mathcal{D}}^2 \mathcal{D}^2 = 16 \square \mathcal{D}^2, \quad -\mathcal{D}^\alpha \bar{\mathcal{D}}^2 \mathcal{D}_\alpha = 8 \square - \frac{1}{2} \{\mathcal{D}^2, \bar{\mathcal{D}}^2\}. \quad (3.34)$$

The Fermi–Feynman quadratic form $S_{E,V}^{(2)} = \frac{1}{4g^2} \int d^4x d^4\theta \mathcal{V}^A \square \mathcal{V}^A$ and the constrained identity kernels $\mathbf{1}_\pm := \mathcal{P}_\pm \delta^8(z - z')$ give the exact vector and matter propagators

$$G^{VV,AB} = 2\hbar g^2 \delta^{AB} \square^{-1}, \quad G^{\Phi\tilde{\Phi},AB} = -\hbar g^2 \delta^{AB} \mathcal{P}_+ \delta^8(z - z'), \quad (3.35)$$

the reversed matter orientation using $\mathcal{P}_-$. In momentum space (phase e^{ipx} , all momenta incoming, $\square \mapsto -p^2$; $\theta_{12} := \theta_1 - \theta_2$):

$$\begin{aligned} \langle \mathcal{V}^A(p, \theta_1) \mathcal{V}^B(p', \theta_2) \rangle &= -(2\pi)^4 \delta^4(p + p') \frac{2\hbar g^2 \delta^{AB}}{p^2} \delta^4(\theta_{12}), \\ \langle \Phi^A(p, \theta_1) \tilde{\Phi}^B(p', \theta_2) \rangle &= (2\pi)^4 \delta^4(p + p') \frac{\hbar g^2 \delta^{AB}}{16p^2} \bar{\mathcal{D}}_1^2 \mathcal{D}_1^2 \delta^4(\theta_{12}). \end{aligned} \quad (3.36)$$

The spinor-derivative algebra collapses products of matter propagators at coincident superspace points through the loop-saturation identity

$$\delta^4(\theta_{12}) \mathcal{D}^2 \bar{\mathcal{D}}^2 \delta^4(\theta_{12}) = 16 \delta^4(\theta_{12}). \quad (3.37)$$

The interaction vertices are ordered superfield words; we abbreviate $\int_8 := \int d^4x d^4\theta$ and $\int_\pm$ for the chiral/antichiral integrals. Expanding (3.3) with the exact exponential derivatives

$$e^{-\mathcal{V}} \mathcal{D}_\alpha e^{\mathcal{V}} = \sum_{p,q \geq 0} \frac{(-1)^p}{p! q! (p+q+1)} \mathcal{V}^p (\mathcal{D}_\alpha \mathcal{V}) \mathcal{V}^q, \quad e^{\mathcal{V}} \bar{\mathcal{D}}_{\dot{\alpha}} e^{-\mathcal{V}} = \sum_{p,q \geq 0} \frac{(-1)^{q+1}}{p! q! (p+q+1)} \mathcal{V}^p (\bar{\mathcal{D}}_{\dot{\alpha}} \mathcal{V}) \mathcal{V}^q, \quad (3.38)$$

the pure-gauge words are

$$S_{E,+}^g = - \sum_{p,q,r,s \geq 0} \frac{(-1)^{p+r} f_{AB}}{256 p! q! r! s! (p+q+1)(r+s+1)} \int_+ [\bar{\mathcal{D}}^2 (\mathcal{V}^p \mathcal{D}^\alpha \mathcal{V} \mathcal{V}^q)]^A [\bar{\mathcal{D}}^2 (\mathcal{V}^r \mathcal{D}_\alpha \mathcal{V} \mathcal{V}^s)]^B, \quad (3.39)$$

the antichiral words carrying $(-1)^{q+s} \tilde{f}_{AB}$ with $\mathcal{D} \leftrightarrow \bar{\mathcal{D}}$; at cubic order this gives the four words with coefficients $\pm f_{AB}/512$. The matter–gauge chain is

$$S_E^{\text{mat}} = -\frac{1}{g^2} \delta_{AB} \sum_{r=1}^{m_N} \sum_{n \geq 0} \frac{1}{n!} \int_8 \tilde{\Phi}_r^A \mathcal{V}^{A_1} \dots \mathcal{V}^{A_n} (T_{A_1} \dots T_{A_n})^B{}_C \Phi_r^C, \quad (3.40)$$

with $(T_{A_1} \dots T_{A_n})^B{}_C = i^n f_{A_1 D_1}{}^B f_{A_2 D_2}{}^D \dots f_{A_n C}{}^{D_{n-1}}$ in the adjoint representation; the labeled-leg vertex carries the color word symmetrized over its n legs, and the cubic vertex is $-\frac{i}{g^2} f_{ABC} \int_8 \tilde{\Phi}^A \mathcal{V}^B \Phi^C$. The $\mathcal{N} = 4$ superpotential contributes the chiral and antichiral cubic primitives

$$\mathcal{V}_{\Phi\Phi\Phi} = \mathcal{V}_{\tilde{\Phi}\tilde{\Phi}\tilde{\Phi}} = \frac{\sqrt{2}}{g^2} \varepsilon_{rst} f^{ABC}. \quad (3.41)$$

The ghost sector is generated by inserting (3.22) into

$$S^{\text{FP}} = \frac{1}{4} \int_+ \bar{c}_+ \bar{\mathcal{D}}^2 (s\mathcal{V}) + \frac{1}{4} \int_- \bar{c}_- \mathcal{D}^2 (s\mathcal{V}), \quad S_{\text{FP}}^{(2)} = \frac{i}{4} \int_+ \bar{c}_+ \bar{\mathcal{D}}^2 \tilde{c} - \frac{i}{4} \int_- \bar{c}_- \mathcal{D}^2 c, \quad (3.42)$$

with ghost–ghost–vector words of coefficients $\frac{i}{4} \frac{B_{p+q}(-1)^q}{p! q!}$ (for $\tilde{c}$) and $-\frac{i}{4} \frac{B_{p+q}(-1)^p}{p! q!}$ (for c), the lowest being the $B_1 = -\frac{1}{2}$ word $s\mathcal{V}|_{(1)} = -\frac{i}{2} [\mathcal{V}, c + \tilde{c}]$. Supergraph weights follow the standard rule: a factor τ per vertex and $-\tau^{-1}$ per propagator, with $\tau = i/\hbar$ Lorentzian and $\tau = -1/\hbar$ Euclidean, graded permutation signs for labeled legs, and division by the automorphism group of the labeled graph.

3.7 Regularization: dimensional reduction

Loop integrations are regularized by dimensional reduction [5]: loop momenta are d -dimensional, $d = 4 - 2\epsilon$, while the spinor algebra, the superspace derivatives and all numerator spin words remain four-dimensional. Two metric representations coexist and must not be confused. The loop-momentum split is

$$\delta_4^{mn} = \widehat{\delta}^{mn} + \check{\delta}^{mn}, \quad \widehat{\delta}^m_m = d, \quad \check{\delta}^m_m = 4 - d = 2\epsilon, \quad \widehat{\delta}^m_r \check{\delta}^r_n = 0, \quad (3.43)$$

with purely hatted loop momenta, $\ell^m = \widehat{\delta}^m_n \ell^n$; the numerator spinor/vector split is

$$\delta_d^{MN} = \bar{\delta}^{MN} + \widetilde{\delta}^{MN}, \quad \text{tr } \bar{\delta} = 4, \quad \text{tr } \widetilde{\delta} = d - 4 = -2\epsilon. \quad (3.44)$$

The regulator enters the computation through a single axiom — finite superspace spin words are contracted with the four-dimensional metric $\bar{\delta}$, while propagator inverses use the d -dimensional metric δ_d — and through the evanescent scalar

$$\mu_\ell^2 := \bar{\ell}^2 - \ell_d^2 = -\widetilde{\ell}^2, \quad (3.45)$$

the squared length of the loop momentum along the -2ϵ evanescent directions. Fourier and measure conventions are

$$f(x) = \int \frac{d^4 p}{(2\pi)^4} e^{ipx} f(p), \quad \square \mapsto -p^2, \quad \int_\ell := \mu^{2\epsilon} \int \frac{d^d \ell}{(2\pi)^d}, \quad (3.46)$$

with all external momenta taken incoming and the superspace delta functions $\delta^4(\theta_{12})$ of Section 3.6.

Remark 3.5. Dimensional reduction is the only scheme-sensitive input of the one-loop computation: the single axiom above, together with the master integrals quoted in Section 5, is the complete regulator data used. Whether evanescent operators are required beyond this axiom is a question about higher-order structure and is deferred to the discussion of Section 4.

4 The superspace Feynman-diagram derivation

This section contains the core computation of the paper: the one-loop anomalous Schwinger–Dyson (Ward) identity for the BPS letters of Euclidean $\mathcal{N} = 4$ super-Yang–Mills theory, quantized in $\mathcal{N} = 1$ superspace [3, 2] and regularized by dimensional reduction (DRED) [5]. We define the letter alphabet (Section 4.1), reduce the tree-level Schwinger–Dyson identity to Euler operators (Section 4.2), isolate the unique regulated cutting failure (Section 4.3), derive the selection rules that organize the eighty-one ordered channels (Section 4.4), compute the seed channel (b, b) in full (Section 4.5), state the complete table of nonzero channels (Section 4.6), and close with the holomorphic-derivative tower kernel (Section 4.7).

4.1 The letters

We work in the Euclidean plus/minus spin frame of Section 2, with gauge-covariant superspace derivatives whose frame components are $\nabla_\pm$ and the dotted components $\mathcal{D}_{+\dot{\alpha}}$. In Euclidean signature the chiral and antichiral fields are independent: no conjugation relates Φ_r to $\widetilde{\Phi}^r$. The semi-chiral fermionic derivative ∇_- is a frame component of the gauge-covariant spinor derivative; it is the operator whose action on composite letters we compute.

Definition 4.1 (BPS letters). The *letters* are the gauge-covariant single-trace composites

$$b := -\frac{i}{\sqrt{2}} \nabla_+ \mathcal{W}_+, \quad \beta_r := \frac{1}{\sqrt{2}} \nabla_+ \Phi_r, \quad \gamma^r := \widetilde{\Phi}^r, \quad \partial_{\dot{\alpha}c} := i \widetilde{\mathcal{W}}_{\dot{\alpha}}, \quad r = 1, 2, 3, \quad \dot{\alpha} = \dot{1}, \dot{2}, \quad (4.1)$$

built from the chiral gaugino field strength $\mathcal{W}_+$, the three adjoint chiral superfields Φ_r , their independent Euclidean antichiral partners $\tilde{\Phi}^r$, and the antichiral field strength $\widetilde{\mathcal{W}}_{\dot{\alpha}}$ of Section 2. These are the Euclidean avatars of the 1/16-BPS letters of the holomorphic twist [1].

The letters carry three gradings, summarized here and used throughout:

letter	definition	Grassmann parity	$SU(3)$ flavor	twisted dimension
b	$-\frac{i}{\sqrt{2}} \nabla_+ \mathcal{W}_+$	0	1	2
β_r	$\frac{1}{\sqrt{2}} \nabla_+ \Phi_r$	1	3	$\frac{3}{2}$
γ^r	$\tilde{\Phi}^r$	0	$\bar{3}$	1
$\partial_{\dot{\alpha}} c$	$i \widetilde{\mathcal{W}}_{\dot{\alpha}}$	1	1	$\frac{3}{2}$

The holomorphic (dotted) covariant derivatives are

$$\partial_{\dot{\alpha}} := \mathcal{D}_{+\dot{\alpha}}, \quad \partial^{\dot{\alpha}} := \epsilon^{\dot{\alpha}\dot{\beta}} \partial_{\dot{\beta}}, \quad \epsilon^{i\dot{2}} = +1, \quad \epsilon_{i\dot{2}} = -1, \quad (4.2)$$

of twisted dimension $[\partial_{\dot{\alpha}}] = 1$, and the elementary output bilinear of the anomaly is the ordered dotted contraction

$$\langle X, Y \rangle := (\partial_{\dot{\alpha}} X)(\partial^{\dot{\alpha}} Y). \quad (4.3)$$

For two odd letters the dotted spinor contraction is symmetric under exchange of the two entries; no symmetry property is needed for mixed-parity pairs.

The alphabet contains nine letters — b (one), β_r (three), γ^r (three), $\partial_{\dot{\alpha}} c$ (two) — and we study the $9 \times 9 = 81$ ordered composite insertions $\nabla_-(L_i^A L_j^B)$. Reversed orderings are independent: (L_i, L_j) and (L_j, L_i) are separate channels, never quotiented. The anomaly operator itself carries twisted dimension $+\frac{1}{2}$, so both sides of the master identity below are homogeneous of dimension $[L_i] + [L_j] + \frac{1}{2}$.

Remark 4.2 (The Q_0 correspondence). The classical supercharge of the BPS sector is identified with the semi-chiral derivative by

$$Q_0 \longleftrightarrow -\frac{1}{2} \nabla_-. \quad (4.4)$$

The tree-level contact terms of the Schwinger–Dyson family are exactly the classical bracket (the Q_0 sector); they are computed and removed first. The one-loop remainder derived below is the one-loop correction to the Q_0 Ward identity. The correspondence (4.4) is used again in Section 7.

4.2 The Schwinger–Dyson identity and the Euler-operator reduction

The regulated path integral obeys a local Schwinger–Dyson identity. Its tree content in the letter sector is a reduction to Euler operators — the functional equation-of-motion operators of the vector and of the antichiral matter fields. Writing $(X \times Y)^A := f^{ABC} X^B Y^C$ for the adjoint commutator, the two Euler operators are

$$\frac{\delta S}{\delta \mathcal{V}} = \nabla^a \mathcal{W}_a - 2i (\Phi_s \times \tilde{\Phi}^s), \quad \frac{\delta S}{\delta \tilde{\Phi}^r} = -\frac{1}{4} \nabla^2 \Phi_r - \frac{1}{\sqrt{2}} \varepsilon_{rst} (\gamma^s \times \gamma^t), \quad (4.5)$$

with a sum over the flavor index s understood, and with the overall g^{-2} of the absorbed frame of Section 3 stripped from the Euler operators throughout this section. Using $\nabla^2 = 2\nabla_- \nabla_+$, the tree-level Schwinger–Dyson reduction of the letters is the exact system

$$\boxed{\begin{aligned} \nabla_- b &= \frac{i}{\sqrt{2}} \nabla_+ \frac{\delta S}{\delta \mathcal{V}} - 2 (\beta_s \times \gamma^s), \\ \nabla_- \beta_r &= -\sqrt{2} \frac{\delta S}{\delta \tilde{\Phi}^r} - \varepsilon_{rst} (\gamma^s \times \gamma^t), \\ \nabla_- \gamma^r &= 0, \quad \nabla_- (\partial_{\dot{\alpha}} c) = 0. \end{aligned}} \quad (4.6)$$

Every ordered pair is then reduced by the graded Leibniz rule

$$\nabla_-(L_i L_j) = (\nabla_- L_i) L_j + (-1)^{|L_i|} L_i (\nabla_- L_j). \quad (4.7)$$

Graphically, an Euler-operator insertion attached to a propagator collapses it, $KK^{-1} = \delta$: this is the *cutting rule*. The surviving tree contact terms are precisely the classical bracket of Remark 4.2. The one-loop content of the same identity is the master result of this paper:

Proposition 4.3 (One-loop anomalous Ward identity). *For each of the eighty-one ordered pairs of letters,*

$$\nabla_-(L_i^A L_j^B) \Big|_{1\text{-loop}} = \frac{\hbar g^2}{16\pi^2} f^{ACD} f^{BCE} \Delta(L_i, L_j)^{DE}, \quad (4.8)$$

where $\Delta(L_i, L_j)$ are the fixed letter bilinears collected in Section 4.6; Δ vanishes for 52 of the 81 ordered pairs.

Remark 4.4 (Classical/quantum separation). If a regulator preserved simultaneously (i) the equation-of-motion/variation correspondence, (ii) the cutting rule $KK^{-1} = \delta$, and (iii) four-dimensional D -algebra, the complete Schwinger–Dyson graph family would cancel and no anomaly could exist. DRED preserves (i) and (iii) but deforms (ii): the quantum remainder is carried solely by the evanescent insertions of Section 4.3.

4.3 The cutting mechanism

The two metric splits of DRED

Dimensional reduction keeps $d = 4 - 2\epsilon$ while splitting the four-dimensional metric and the d -dimensional metric as

$$\delta_4^{mn} = \widehat{\delta}^{mn} + \check{\delta}^{mn}, \quad \widehat{\delta}_m^m = d, \quad \check{\delta}_m^m = 4 - d = 2\epsilon, \quad \widehat{\delta}_r^m \check{\delta}_n^r = 0, \quad (4.9)$$

the quasi-four-dimensional split, and the scalar dimension-shift split

$$\delta_d^{MN} = \bar{\delta}^{MN} + \widetilde{\delta}^{MN}, \quad \text{tr } \bar{\delta} = 4, \quad \text{tr } \widetilde{\delta} = d - 4 = -2\epsilon, \quad \ell_d^2 = \bar{\ell}^2 + \widetilde{\ell}^2. \quad (4.10)$$

The loop momentum has only hatted components, $\ell^m = \widehat{\delta}_n^m \ell^n$, and the measure is $\mu^{2\epsilon} \int d^d \ell / (2\pi)^d$. The evanescent loop-momentum square is

$$\boxed{\mu_\ell^2 := \bar{\ell}^2 - \ell_d^2 = -\widetilde{\ell}^2} \quad (4.11)$$

and the single regulator axiom of the computation is

$$\boxed{\text{finite superspace spin words use } \bar{\delta}, \quad \text{propagator inverses use } \delta_d.} \quad (4.12)$$

The spin algebra is four-dimensional in this representation,

$$\sigma_m \bar{\sigma}_n + \sigma_n \bar{\sigma}_m = 2\bar{\delta}_{mn} \mathbf{1}_2. \quad (4.13)$$

Remark 4.5 (The two splits are never contracted with each other). Combining the two representations (4.9) and (4.10) before the Schwinger family is summed would set the scalar difference (4.11) to zero and erase the anomaly. Our Fourier convention is $X(x) = \int \frac{d^d p}{(2\pi)^d} e^{ip \cdot x} X(p)$ with all momenta incoming, so $\square \mapsto -p^2$.

The exact cutting-failure identity

On an internal edge e the momentum is $r_e = \ell + Q_e$, and the external shifts Q_e carry no evanescent components, so $\bar{r}_e^2 = r_{e,d}^2 + \mu_\ell^2$ and

$$\frac{\bar{r}_e^2}{r_{e,d}^2} = 1 + \frac{\mu_\ell^2}{r_{e,d}^2}. \quad (4.14)$$

The first term is exactly the Schwinger cut. For a triangle with denominators $D_i = r_{i,d}^2$ one therefore has, for *every marked line e of every one-loop graph*, the exact cutting-failure identity

$$\boxed{\frac{\bar{r}_e^2}{D_0 D_1 D_2} - \frac{1}{\prod_{j \neq e} D_j} = \frac{\mu_\ell^2}{D_0 D_1 D_2}} \quad (4.15)$$

— one universal insertion, with no graph-specific $(4-d)$ factor. The entire anomaly is the sum of these remainders. Replacing $\bar{r}_e^2$ by $r_{e,d}^2$ before the subtraction makes the expression identically zero and removes the anomaly.

The evanescent master integral

Finiteness and locality of the anomaly follow from the master integral

$$\boxed{\lim_{\epsilon \rightarrow 0} \mu^{2\epsilon} \int \frac{d^{4-2\epsilon} \ell}{(2\pi)^{4-2\epsilon}} \frac{\mu_\ell^2}{(\ell_d^2 + \Delta)^3} = \frac{1}{32\pi^2}} \quad (4.16)$$

which is independent of the Feynman-parameter combination Δ . The proof uses two ingredients. First, by d -dimensional rotational invariance inside a rotationally invariant integral,

$$\int \tilde{\ell}^2 f(\ell^2) = \frac{d-4}{d} \int \ell^2 f(\ell^2). \quad (4.17)$$

Second, the scalar master family

$$J_n(\Delta) := \mu^{2\epsilon} \int \frac{d^d r}{(2\pi)^d} \frac{1}{(r^2 + \Delta)^n} = \frac{\mu^{2\epsilon}}{(4\pi)^{d/2}} \frac{\Gamma(n-d/2)}{\Gamma(n)} \Delta^{d/2-n} \quad (4.18)$$

obeys

$$\Delta J_3 = \frac{\epsilon}{2} J_2, \quad \text{Res}_{\epsilon=0} J_2 = \frac{1}{16\pi^2}, \quad J_2 - \Delta J_3 = \left(1 - \frac{\epsilon}{2}\right) J_2. \quad (4.19)$$

Combining (4.11), (4.17) and (4.19),

$$\int_\ell \frac{\mu_\ell^2}{(\ell_d^2 + \Delta)^3} = -\frac{d-4}{d} (J_2 - \Delta J_3) = \frac{2\epsilon}{4-2\epsilon} \cdot \frac{4-2\epsilon}{4} J_2 = \frac{\epsilon}{2} J_2 \xrightarrow{\epsilon \rightarrow 0} \frac{1}{2} \cdot \frac{1}{16\pi^2} = \frac{1}{32\pi^2}, \quad (4.20)$$

using $\epsilon \Gamma(\epsilon) \rightarrow 1$. The would-be nonlocal Δ -dependence enters only at $O(\epsilon)$ through $\Delta^{-\epsilon}$, so the anomaly is a polynomial in external momenta: a local operator.

Remark 4.6 (Pole bookkeeping as a cross-check principle). For each channel the isolated parent graphs are UV divergent; the complete Schwinger–Dyson family is finite. The $1/\epsilon$ poles cancel between the parent tensor reduction, which involves the physical metric $\hat{\delta}$, and the Schwinger contact rows, which involve the four-dimensional metric δ_4 , leaving $\hat{\delta} - \delta_4 = -\check{\delta}$: a pure evanescent numerator whose trace supplies the finite answer. Equivalently, every anomaly factorizes as (evanescent trace $\propto 2\epsilon$) $\times$ (pole residue $1/(16\pi^2\epsilon)$). Any error in a single vertex factor or sign would break this cancellation, so the finiteness of each channel is itself a consistency proof of its factor chain.

4.4 Selection rules: the census of the eighty-one ordered pairs

Before any loop integral, conservation laws restrict the possible uncontracted outputs. The gradings that survive the construction are:

- (i) $SU(3)$ flavor: b and $\partial_{\dot{\alpha}}c$ are singlets, $\beta_r \in \mathbf{3}$, $\gamma^r \in \bar{\mathbf{3}}$;
- (ii) Grassmann parity: $|b| = |\gamma| = 0$, $|\beta_r| = |\partial_{\dot{\alpha}}c| = 1$;
- (iii) twisted dimension: $[\gamma] = 1$, $[\beta] = [\partial c] = \frac{3}{2}$, $[b] = 2$, $[\partial_{\dot{\alpha}}] = 1$, with the anomaly operator carrying $+\frac{1}{2}$;
- (iv) the output structure $\langle X, Y \rangle$ of equation (4.3) — exactly one holomorphic derivative on each of two letters — forced by the σ -chain trace of Section 4.5.

Checking each ordered family against these gradings and against the cubic-vertex spectator structure yields exactly

$$29 \text{ nonzero} = 1_{(b,b)} + 6_{(b,\beta),(\beta,b)} + 6_{(b,\gamma),(\gamma,b)} + 4_{(b,\partial c),(\partial c,b)} + 6_{(\beta,\gamma)} \delta\text{-diagonal} + 6_{(\beta,\beta)} \varepsilon\text{-offdiagonal}. \quad (4.21)$$

The remaining 52 ordered pairs vanish exactly, in four structural classes:

1. **25 no-descendant pairs.** Since $\nabla_- \gamma = \nabla_- (\partial c) = 0$, the Leibniz rule (4.7) makes $\nabla_- (L_i L_j)$ vanish identically when both letters are drawn from $\{\gamma^1, \gamma^2, \gamma^3, \partial_1 c, \partial_2 c\}$ — all $5 \times 5 = 25$ pairs of this set.
2. **12 flavor-off-diagonal** $(\beta_r, \gamma^s)/(\gamma^s, \beta_r)$, $r \neq s$. Exact zeros: the available singlet contraction $\delta_{rs} \langle \partial c, \partial c \rangle$ has no off-diagonal support.
3. **3 diagonal** (β_r, β_r) . Exact zeros: the candidate D -word and the repeated-index ε routes both vanish.
4. **12** $(\beta_r, \partial_{\dot{\alpha}}c)/(\partial_{\dot{\alpha}}c, \beta_r)$. Killed exactly by the charge and cubic-vertex basis analysis.

Remark 4.7 (Three independent routes to the same census). The 29/52 census is obtained independently by (a) the selection-rule analysis above, (b) the explicit evaluation of all eighty-one ordered Schwinger–Dyson families, and (c) the vanishing list of the holomorphic-twist calculation [1] (Section 7). All three routes agree.

Remark 4.8 (Status). The eighty-one-row result — twenty-nine exact nonzero channels and fifty-two exact zeros — is the accepted physical one-loop anomaly sector of the project, with the physical verifier passing all thirty-three checks. An adversarial census review (seventeen candidate structures) remains open; it does not alter the accepted coefficients stated here.

4.5 The seed channel (b, b) in full

We now compute the seed channel (b, b) , the simplest nonzero channel and the scale anchor of the whole table. We fix once and for all an orientation of the two field-strength insertions and a marked internal line. The graph weights below are computed with unit-normalized field-strength insertions $\nabla_+ \mathcal{W}_+$; the letter normalization $b = -(i/\sqrt{2})\nabla_+ \mathcal{W}_+$ is restored at the end.

1. Vertex factors. The two ordered cubic gauge vertices contribute

$$\left(+\frac{ig}{2}\right) \left(-\frac{ig}{2}\right) = \frac{g^2}{4}. \quad (4.22)$$

2. Wick weight. The two equivalent Wick contractions give

$$w_{\text{Wick}} = \frac{1}{2!}(1+1) = 1. \quad (4.23)$$

3. D -algebra weight. The closed-loop mixed D -algebra weight is

$$w_D = \frac{1}{32} \cdot 16 \cdot 2 \cdot 2 = 2, \quad (4.24)$$

where the 16 is the loop-saturation identity $\delta^4(\theta_{12}) \mathcal{D}^2 \bar{\mathcal{D}}^2 \delta^4(\theta_{12}) = 16 \delta^4(\theta_{12})$ of Section 3. The total prefactor before the loop integral is therefore

$$w_{\text{Wick}} \frac{g^2}{4} w_D = 1 \cdot \frac{g^2}{4} \cdot 2 = \frac{g^2}{2}. \quad (4.25)$$

4. Denominators and ordered Feynman parametrization. With $D_0 = \ell^2$, $D_1 = (\ell+p)^2$, $D_2 = (\ell+p+q)^2$,

$$\frac{1}{D_0 D_1 D_2} = 2 \int_{x,y,z \geq 0} dx dy dz \delta(1-x-y-z) \frac{1}{(r^2 + \Delta)^3}, \quad \int_{x,y,z \geq 0} dx dy dz \delta(1-x-y-z) = \frac{1}{2}, \quad (4.26)$$

with r the shifted loop momentum and Δ the Feynman-parameter combination of external momenta.

5. Tensor reduction: the exact $\frac{1}{4}$. Hatted rotational reduction and (4.19) give

$$\int_r \frac{r_m r_n}{(r^2 + \Delta)^3} = \frac{\hat{\delta}_{mn}}{d} (J_2 - \Delta J_3) = \frac{\hat{\delta}_{mn}}{4} J_2, \quad (4.27)$$

the $\frac{1}{4}$ being exact since $(1 - \frac{\epsilon}{2})/(4 - 2\epsilon) = \frac{1}{4}$ identically in ϵ . Consequently the pole part of the parent triangle with numerator legs $L_1^m L_2^n$ is

$$\text{Pole}_{\epsilon=0} \int_\ell \frac{L_1^m L_2^n}{D_0 D_1 D_2} = \frac{\hat{\delta}^{mn}}{16\pi^2 \epsilon}. \quad (4.28)$$

6. Parent triangle pole. Assembling the weights (4.25) with the spinor chain $\sigma_m \bar{\sigma}_\rho \sigma_n$,

$$\Gamma_T = \frac{\hbar g^2}{32\pi^2 \epsilon} f^{ACD} f^{BCE} \hat{\delta}^{mn} \sigma_m \bar{\sigma}_\rho \sigma_n p^\rho. \quad (4.29)$$

7. Schwinger contact rows. Each contact row carries the sign chain

$$\left(-\frac{1}{4}\right)_{\text{cut}} (-1)_{\text{endpoint}} \left(\frac{1}{2}\right)_{\nabla_- \nabla_+ = \nabla^2/2} (2)_{\text{mixed}} (2)_{\text{Wick}} \left(-\frac{1}{2}\right)_{p_+} = -\frac{1}{4}, \quad (4.30)$$

so a single row gives

$$\Gamma_{C,j} = -\frac{1}{4} \cdot \frac{\hbar g^2}{2} \cdot \frac{1}{16\pi^2 \epsilon} f^{ACD} f^{BCE} \delta_4^{mn} \sigma_m \bar{\sigma}_\rho \sigma_n p^\rho = -\frac{\hbar g^2}{128\pi^2 \epsilon} f^{ACD} f^{BCE} \delta_4^{mn} \sigma_m \bar{\sigma}_\rho \sigma_n p^\rho, \quad (4.31)$$

and the four occurrence-resolved rows sum to

$$\Gamma_C = \sum_{j=1}^4 \Gamma_{C,j} = -\frac{\hbar g^2}{32\pi^2 \epsilon} f^{ACD} f^{BCE} \delta_4^{mn} \sigma_m \bar{\sigma}_\rho \sigma_n p^\rho. \quad (4.32)$$

8. Pole cancellation. The four-dimensional metric parts of (4.29) and (4.32) cancel exactly: since $\widehat{\delta}^{mn} - \delta_4^{mn} = -\check{\delta}^{mn}$,

$$\Gamma_T + \Gamma_C = -\frac{\hbar g^2}{32\pi^2\epsilon} f^{ACD} f^{BCE} \check{\delta}^{mn} \sigma_m \bar{\sigma}_\rho \sigma_n p^\rho, \quad (4.33)$$

a pure evanescent numerator, precisely as in the bookkeeping principle of Remark 4.6.

9. The -2ϵ sigma-trace. The external momentum is hatted, so contracting with p^ρ first and using $\check{\delta} p = 0$ and $\check{\delta}^{mn} \bar{\sigma}_m \sigma_n = \check{\delta}^m_m = 2\epsilon$,

$$p^\rho \check{\delta}^{mn} \sigma_m \bar{\sigma}_\rho \sigma_n = 2p^\rho \check{\delta}_\rho^n \sigma_n - p^\rho \sigma_\rho \check{\delta}^{mn} \bar{\sigma}_m \sigma_n = 0 - (\check{\delta}^m_m) p^\rho \sigma_\rho = -2\epsilon p^\rho \sigma_\rho. \quad (4.34)$$

10. Final finite local result. Substituting (4.34) into (4.33),

$$\Gamma_T + \Gamma_C = -\frac{\hbar g^2}{32\pi^2\epsilon} (-2\epsilon) f^{ACD} f^{BCE} \sigma_\rho p^\rho = \frac{\hbar g^2}{16\pi^2} f^{ACD} f^{BCE} \sigma \cdot p. \quad (4.35)$$

The σ -chain trace forces the position-space output to be a dotted bilinear of the form (4.3): this is the output structure quoted as grading (iv) in Section 4.4. Restoring the letter normalization — the two insertions contribute a factor $(-i/\sqrt{2})^2 = -\frac{1}{2}$ relative to the unit field-strength normalization used above, while the output bilinears of the seed channel contribute $\sqrt{2}$ in the alphabet of Definition 4.1 — the position-space identity reads

$$\nabla_-(b^A b^B) = -\frac{\hbar g^2}{16\sqrt{2}\pi^2} f^{ACD} f^{BCE} \left\{ \langle \partial c, b \rangle - \langle b, \partial c \rangle + \sum_{r=1}^3 (\langle \beta_r, \gamma^r \rangle - \langle \gamma^r, \beta_r \rangle) \right\}^{DE}.$$

(4.36)

Remark 4.9 (Independent machine cross-checks). The factor chain of this subsection was replayed externally on 269 external slot assignments with zero failures, and on the full color mask (9216 entries, with a sparse replay of 2048) with zero equality failures.

4.6 The complete channel table

We now state all seven nonzero families of the master identity (4.8). The letter definitions (4.1) induce a uniform rescaling between the unit-normalized field-strength composites used in the graph computations and the letters of Definition 4.1. The bookkeeping is immediate: in the seed channel the two insertions contribute $(-i/\sqrt{2})^2 = -\frac{1}{2}$ and the output bilinears contribute $\sqrt{2}$, giving the net prefactor $-\frac{1}{2} \cdot \sqrt{2} \cdot \frac{\hbar g^2}{16\pi^2} = -\frac{\hbar g^2}{16\sqrt{2}\pi^2}$ of equation (4.36). The same arithmetic, applied family by family, yields the common prefactor $-\hbar g^2/(16\sqrt{2}\pi^2)$ for every family, including the $(b, \beta)/(\beta, b)$ family, whose output word is fixed by the residual-supersymmetry closure discussed below. With $\langle X, Y \rangle$ as in (4.3) and $\partial_{\check{\alpha}}(\partial c)$ the holomorphic derivative of the letter

∂c , the complete set of nonzero channels is

$$\nabla_{-}(b^A b^B) = -\frac{\hbar g^2}{16\sqrt{2}\pi^2} f^{ACD} f^{BCE} \left\{ \langle \partial c, b \rangle - \langle b, \partial c \rangle + \sum_r (\langle \beta_r, \gamma^r \rangle - \langle \gamma^r, \beta_r \rangle) \right\}^{DE}, \quad (4.37)$$

$$\nabla_{-}(b^A \gamma^{rB}), \quad \nabla_{-}(\gamma^{rA} b^B) = -\frac{\hbar g^2}{16\sqrt{2}\pi^2} f^{ACD} f^{BCE} \left\{ \langle \partial c, \gamma^r \rangle - \langle \gamma^r, \partial c \rangle \right\}^{DE}, \quad (4.38)$$

$$\nabla_{-}(\beta_r^A \gamma^{sB}), \quad \nabla_{-}(\gamma^{sA} \beta_r^B) = -\frac{\hbar g^2}{16\sqrt{2}\pi^2} f^{ACD} f^{BCE} \delta_{rs} \langle \partial c, \partial c \rangle^{DE}, \quad (4.39)$$

$$\nabla_{-}(\beta_r^A \beta_s^B) = -\frac{\hbar g^2}{16\sqrt{2}\pi^2} f^{ACD} f^{BCE} \varepsilon_{rst} \left\{ \langle \partial c, \gamma^t \rangle - \langle \gamma^t, \partial c \rangle \right\}^{DE}, \quad (4.40)$$

$$\nabla_{-}(b^A \beta_r^B), \quad \nabla_{-}(\beta_r^A b^B) = -\frac{\hbar g^2}{16\sqrt{2}\pi^2} f^{ACD} f^{BCE} \left\{ \langle \beta_r, \partial c \rangle + \langle \partial c, \beta_r \rangle + \varepsilon_{rst} \langle \gamma^s, \gamma^t \rangle \right\}^{DE}, \quad (4.41)$$

$$\nabla_{-}(b^A \partial_{\dot{\alpha}} c^B) = -\frac{\hbar g^2}{16\sqrt{2}\pi^2} f^{ACD} f^{BCE} \left\{ \frac{1}{3} \langle \partial_{\dot{\alpha}}(\partial c), \partial c \rangle + \frac{2}{3} \langle \partial c, \partial_{\dot{\alpha}}(\partial c) \rangle \right\}^{DE}, \quad (4.42)$$

$$\nabla_{-}(\partial_{\dot{\alpha}} c^A b^B) = -\frac{\hbar g^2}{16\sqrt{2}\pi^2} f^{ACD} f^{BCE} \left\{ \frac{2}{3} \langle \partial_{\dot{\alpha}}(\partial c), \partial c \rangle + \frac{1}{3} \langle \partial c, \partial_{\dot{\alpha}}(\partial c) \rangle \right\}^{DE}. \quad (4.43)$$

The free dotted index in (4.42) and (4.43) is the letter index of $\partial_{\dot{\alpha}} c$ on the left and the extra holomorphic derivative on the right; the weights $(\frac{1}{3}, \frac{2}{3})$ are the $(m, n) = (1, 0)$ instance of the tower kernel of Section 4.7. Together with the fifty-two zeros of Section 4.4, the rows (4.37)–(4.43) exhaust the eighty-one ordered channels: $1 + 6 + 6 + 4 + 6 + 6 = 29$ nonzero, as in (4.21).

Residual-supersymmetry closure of the $(b, \beta)/(\beta, b)$ channels

The family (4.41) is the only one whose output word is not read off a cut parent graph directly. The raw graph carrier of this family contains an equation-of-motion (total-derivative) layer; the physical answer is the unique residual-supersymmetry-closed vector obtained after removing that layer and adding a finite normal-product (scheme) term, with the remaining one-parameter freedom anchored to the (b, b) scale. The closure works as follows. The residual supercharges of the bottom-letter sector, $q_r := \frac{1}{\sqrt{2}} Q_{E,+}^r$, act on the letters by

$$q_r b = 0, \quad q_r \beta_s = \delta_{rs} b, \quad q_r \gamma^s = -\varepsilon_{rst} \beta_t, \quad q_r (\partial_{\dot{\alpha}} c) = \partial_{\dot{\alpha}} \gamma^r. \quad (4.44)$$

At twisted dimension $\frac{9}{2}$, the space of odd $SU(3)$ -singlet letter bilinears annihilated by all q_r is one-dimensional, spanned by the seed output word

$$\left\{ \langle \partial c, b \rangle - \langle b, \partial c \rangle + \sum_r (\langle \beta_r, \gamma^r \rangle - \langle \gamma^r, \beta_r \rangle) \right\}^{DE} \quad (4.45)$$

of (4.37). The (b, β_r) output word is its unique triplet q -primitive:

$$q_s \left\{ \langle \beta_r, \partial c \rangle + \langle \partial c, \beta_r \rangle + \varepsilon_{rtu} \langle \gamma^t, \gamma^u \rangle \right\} = -\delta_{sr} \left\{ \langle \partial c, b \rangle - \langle b, \partial c \rangle + \sum_{r'} (\langle \beta_{r'}, \gamma^{r'} \rangle - \langle \gamma^{r'}, \beta_{r'} \rangle) \right\}. \quad (4.46)$$

Equivalently, in the ordered basis $(\langle \partial c, \beta_r \rangle, \langle \beta_r, \partial c \rangle, \langle \gamma^s, \gamma^t \rangle, \langle \gamma^t, \gamma^s \rangle)$ with $\varepsilon_{rst} = +1$, the three Ward equations $q_s[\cdot] = 0$ ($s = 1, 2, 3$) cut out the one-dimensional ray spanned by $(1, 1, 1, -1)$, and anchoring this ray to the (b, b) scale through (4.46) fixes the unit normalization displayed

in (4.41). Since $q_r(b^A b^B) = 0$, the seed channel's own forward orbit is only itself: the physical families were each computed directly, and the closure is a cross-check that fixes the (b, β) normalization.

Remark 4.10 (The finite shift is a scheme term, not an anomaly graph). The passage from the raw graph carrier to the renormalized coefficient vector of the $(b, \beta)/(\beta, b)$ family removes an equation-of-motion (total-derivative) layer and adds a finite normal-product term. This is a finite scheme adjustment, not an additional anomaly graph: all bare-graph anomaly in every channel comes only from the μ_ℓ^2 cutting failure (4.15). The closure was verified by thirty-one independent finite Ward-identity checks. Through this finite shift and the anchoring to the (b, b) scale, the $(b, \beta)/(\beta, b)$ family lands on the same universal prefactor $-\hbar g^2/(16\sqrt{2}\pi^2)$ as every other nonzero family, with the output word displayed in (4.41).

4.7 The holomorphic-derivative tower

Shifting the second letter of a channel bilinear by a formal vector w — equivalently, resumming all holomorphic derivatives — inserts the leg phase $e^{iw \cdot (aq+bp)}$ into the ordered Feynman parametrization of the triangle. The ordered-simplex moment is

$$\int_{0 < a < b < 1} a^p b^q da db = \frac{1}{(p+1)(p+q+2)}, \quad (4.47)$$

so that

$$2 \int_0^1 db \int_0^b da a^{k+\ell} b^{(m-k)+(n-\ell)} = \frac{2}{(k+\ell+1)(m+n+2)}. \quad (4.48)$$

Restoring the binomial weights of the derivative expansion gives the one-loop holomorphic-derivative kernel

$$\boxed{K_{m,n;k,\ell} = \frac{2 \binom{m}{k} \binom{n}{\ell}}{(m+n+2)(k+\ell+1)}} \quad (m, n \in \mathbb{Z}_{\geq 0}, 0 \leq k \leq m, 0 \leq \ell \leq n). \quad (4.49)$$

Remark 4.11 (The overall factor 2 is $\Gamma(3)$). The numerator factor 2 in (4.49) is $\Gamma(3)$ from combining the three denominators, $\frac{1}{D_0 D_1 D_2} = 2 \int_{\Delta_2} \frac{1}{(r^2 + \Delta)^3}$; it is not an orientation multiplicity and not a second loop orientation.

Because the $\partial_{\dot{\alpha}}$ are gauge-covariant and do not commute, derivative monomials are understood as shuffle-symmetrized averages over the (m, n) orderings of $\partial_1^m \partial_2^n$. The full kernel operator on letter pairs is

$$\mathcal{K}_{m,n}(f^D, g^E) = \sum_{k=0}^m \sum_{\ell=0}^n \frac{2 \binom{m}{k} \binom{n}{\ell}}{(m+n+2)(k+\ell+1)} (\partial_1^k \partial_2^\ell \partial_{\dot{\alpha}} f)^D (\partial_1^{m-k} \partial_2^{n-\ell} \partial_{\dot{\alpha}} g)^E. \quad (4.50)$$

Its first instances are

$$\mathcal{K}_{0,0} = \langle f, g \rangle, \quad \mathcal{K}_{1,0} = \frac{1}{3} \langle \partial_1 f, g \rangle + \frac{2}{3} \langle f, \partial_1 g \rangle, \quad (4.51)$$

$$\mathcal{K}_{2,0} = \frac{1}{6} \langle \partial_1^2 f, g \rangle + \frac{1}{2} \langle \partial_1 f, \partial_1 g \rangle + \frac{1}{2} \langle f, \partial_1^2 g \rangle, \quad (4.52)$$

$$\mathcal{K}_{1,1} = \frac{1}{6} \langle \partial_1 \partial_2 f, g \rangle + \frac{1}{4} \langle \partial_1 f, \partial_2 g \rangle + \frac{1}{4} \langle \partial_2 f, \partial_1 g \rangle + \frac{1}{2} \langle f, \partial_1 \partial_2 g \rangle. \quad (4.53)$$

In particular the $(b, \partial c)$ weights $(\frac{1}{3}, \frac{2}{3})$ of (4.42)–(4.43) are the $(m, n) = (1, 0)$ instance: $K_{1,0;1,0} = \frac{2}{3 \cdot 2} = \frac{1}{3}$ for the derivative on the first letter and $K_{1,0;0,0} = \frac{2}{3 \cdot 1} = \frac{2}{3}$ for the derivative on the second. The comparison of (4.49) with the holomorphic-twist kernel of [1] is the subject of Section 7.

5 The component Feynman-diagram approach

5.1 The program: component quantization of the same action

The one-loop anomalous supersymmetry Ward identity of this paper was derived in Euclidean $\mathcal{N} = 1$ superspace (Section 4). The present section develops the parallel component-field approach: the same Euclidean action of Sections 2 and 3, expanded in component fields and quantized by ordinary Feynman diagrams. An independent component check is desirable for three reasons: the component expansion makes the anomaly directly readable as a bottom-component transformation law; the off-shell formulation carries an off-shell Noether current for the supersymmetry, in whose divergence the anomaly would appear as an Euler-operator insertion, checkable term by term [2, 3]; and ordinary Wick diagrams, regulable by dimensional reduction [5], give a graph-by-graph cross-check of the superspace Schwinger–Dyson derivation, with no superspace D -algebra anywhere in the loop evaluation.

The component content of the theory is the $\mathcal{N} = 1$ vector multiplet (A_m, λ, D) together with three adjoint chiral multiplets (ϕ_r, ψ_r, F_r) , $r = 1, 2, 3$, and their independent Euclidean antichiral partners $(\tilde{\phi}_r, \tilde{\psi}_r, \tilde{F}_r)$ and $\tilde{\lambda}$: tilded and untilded fields are independent integration variables in Euclidean signature and become Lorentzian conjugates only after a real integration cycle is selected. With the color commutator $[X, Y]^A := f^{BCA} X^B Y^C$ and $\kappa_{AB} = \delta_{AB}$ (Section 2), the exact Euclidean off-shell $\mathcal{N} = 4$ component Lagrangian is

$$\mathcal{L}_{E,\text{off}}^{(4)} = \frac{1}{g^2} \delta_{AB} \left[\frac{1}{4} F_{mn}^A F_{mn}^B + \tilde{\lambda}^A \bar{\sigma}_E^m \mathcal{D}_m \lambda^B - \frac{1}{2} D^A D^B + (\mathcal{D}_m \tilde{\phi}_r)^A (\mathcal{D}_m \phi_r)^B + \tilde{\psi}_r^A \bar{\sigma}_E^m \mathcal{D}_m \psi_r^B - \tilde{F}_r^A F_r^B - i D^A [\phi_r, \tilde{\phi}_r] \right. \\ \left. - \frac{\sqrt{2}}{g^2} f^{ABC} (\tilde{\phi}_r^A \psi_r^B \lambda^C + \phi_r^A \tilde{\psi}_r^B \tilde{\lambda}^C) + \frac{1}{\sqrt{2} g^2} \varepsilon_{rst} f^{ABC} (F_r^A \phi_s^B \phi_t^C + \tilde{F}_r^A \tilde{\phi}_s^B \tilde{\phi}_t^C - \phi_t^A \psi_r^B \psi_s^C - \tilde{\phi}_t^A \tilde{\psi}_r^B \tilde{\psi}_s^C) \right], \quad (5.1)$$

plus a topological term that plays no role below and is omitted from the display. Here D^A is the vector-multiplet auxiliary scalar (not a derivative), $\bar{\sigma}_E^m$ are the Euclidean sigma matrices, and repeated flavor indices are summed. Eliminating the auxiliaries through $D = -i[\phi_r, \tilde{\phi}_r]$, $\tilde{F}_r = \frac{1}{\sqrt{2}} \varepsilon_{rst} [\phi_s, \phi_t]$, $F_r = \frac{1}{\sqrt{2}} \varepsilon_{rst} [\tilde{\phi}_s, \tilde{\phi}_t]$ gives the on-shell component action. The two Yukawa families $-\frac{\sqrt{2}}{g^2} f^{ABC} \tilde{\phi}_r^A \psi_r^B \lambda^C$ and $-\frac{\sqrt{2}}{g^2} f^{ABC} \phi_r^A \tilde{\psi}_r^B \tilde{\lambda}^C$ in (5.1) are the only vertices carrying the fields of the anomaly; they are the vertices from which a naive ordinary component derivation would have to be built (Section 5.2).

The component framework also locks the BV–BRST structure exactly. With chiral and antichiral Faddeev–Popov ghost superfields c and $\tilde{c}$, both of ghost number $+1$, the BRST differential acts on the unrestricted vector prepotential $\mathcal{V}$ and the matter superfields as

$$s e^\mathcal{V} = i \tilde{c} e^\mathcal{V} - i e^\mathcal{V} c, \quad s c = i c^2, \quad s \tilde{c} = i \tilde{c}^2, \quad s \Phi_r = i c \Phi_r, \quad s \tilde{\Phi}_r = -i \tilde{\Phi}_r \tilde{c}. \quad (5.2)$$

This differential is nilpotent, $s^2 = 0$. The minimal Batalin–Vilkovisky action obeys the classical master equation and, after the finite-cutoff regulator insertion, the order-by-order quantum master equation [4]. Gauge fixing uses a nonminimal sector with the antighost $\bar{c}$ and the gauge functions $\mathcal{F}_\pm(\mathcal{V}) = -\frac{1}{4} \bar{\mathcal{D}}^2 \mathcal{V}$, $-\frac{1}{4} \mathcal{D}^2 \mathcal{V}$; the Wess–Zumino surface is not imposed before gauge fixing.

Remark 5.1 (Partial scope of the component Feynman rules). Locked exactly are the component actions, the BV–BRST structure, and the all-order primitive graph grammar with labeled-graph weights; this grammar becomes a numerical Feynman rule only after every edge inverse and integration cycle is locked. The Fermi–Feynman gauge-averaged propagators exist only conditionally: the corresponding gauge-fixing kernel contains $\square^{-1}$ and is nonlocal, so an admissible completed perturbative slice is not claimed (status `PASS_EXACT_PARTIAL_SCOPE`). The accepted superspace computation of Section 4 therefore adopts the Fermi–Feynman Gaussian

average as a recorded scheme decision and keeps the Nielsen–Kallosh factor as an external, field-independent measure. Former claims of complete component momentum-space Feynman rules are NOT_ACCEPTED_COMPLETE_FEYNMAN_RULES.

This component approach is one of four mutually independent computations into which the proof program is organized: signature (Euclidean or Lorentzian) times formulation (superspace or component). Each works from the shared frozen conventions and its own derivations only, and results are compared only at the final canonical level. Gating rules order them: the Euclidean component computation begins once the conventions are frozen, the Euclidean superspace one starts after the Euclidean component canonical result is locked, and the two Lorentzian ones start only after a Euclidean bridge is frozen. The accepted component statement below is in fact obtained as the bottom projection of the Euclidean superspace result—explicitly not as an independent component-box derivation.

5.2 The accepted result: bottom projection of the (β, γ) anomaly

Section 4 establishes, for the ordered letter pair (β_r, γ^s) and equally for the reversed ordering (γ^s, β_r) , the exact one-loop identity

$$\nabla_- (\beta_r^A \gamma^s B) \Big|_{1\text{-loop}} = -\frac{\hbar g^2}{16\sqrt{2}\pi^2} \delta_{rs} f^{ACD} f^{BCE} \langle \partial c^D, \partial c^E \rangle, \quad (5.3)$$

where $\langle X, Y \rangle := (\partial_{\dot{\alpha}} X)(\partial^{\dot{\alpha}} Y)$ is the dotted pairing and ∇_- is the anomalous frame component of the gauge-covariant spinor derivative. Its component content is read off by bottom projection. The bottom components of the letters are

$$\beta_r^A \Big| = \psi_{r+}^A, \quad \gamma^s B \Big| = \tilde{\phi}_s^B, \quad \partial_{\dot{\alpha}} c^D \Big| = -\tilde{\lambda}_{\dot{\alpha}}^D, \quad (5.4)$$

where ψ_{r+} is the $+$ -frame bottom component of the r -th matter Weyl fermion, $\tilde{\phi}_s$ the bottom scalar of the antichiral multiplet, and $\tilde{\lambda}$ the dotted gaugino. The component variation Q_- is the bottom action induced by the same locked ∇_- acting on the letters; no independent rephasing is introduced. The dotted pairing projects as

$$\langle \partial c^D, \partial c^E \rangle \Big| = (\partial_{\dot{\alpha}} c^D)(\partial^{\dot{\alpha}} c^E) \Big| = \tilde{\lambda}_{\dot{\alpha}}^D \tilde{\lambda}^{E\dot{\alpha}}. \quad (5.5)$$

Combining (5.3)–(5.5) gives the accepted component formula

$$\boxed{Q_- \left(\psi_{r+}^A \tilde{\phi}_s^B \right) \Big|_{1\text{-loop}} = -\frac{\sqrt{2}\hbar g^2}{32\pi^2} \delta_{rs} f^{ACD} f^{BCE} \tilde{\lambda}_{\dot{\alpha}}^D \tilde{\lambda}^{E\dot{\alpha}}.} \quad (5.6)$$

Remark 5.2 (Status of the boxed formula). Equation (5.6) is an exact projection of the accepted superfield result of Section 4, not an independent component-box derivation. It fixes the former free sign of the component-box proposal to -1 , and the loop order contains exactly one power of $\hbar$.

The dotted gaugino bilinear on the right of (5.6) has a color symmetry that constrains every component realization. With $\epsilon^{12} = +1$, define

$$S^{DE} := \tilde{\lambda}_{\dot{\alpha}}^D \tilde{\lambda}^{E\dot{\alpha}} = \tilde{\lambda}_1^D \tilde{\lambda}_2^E - \tilde{\lambda}_2^D \tilde{\lambda}_1^E, \quad S^{ED} = S^{DE}, \quad (5.7)$$

where the second relation is Grassmann reordering. Consequently, any antisymmetric color coefficient $A_{DE} = -A_{ED}$ obeys

$$A_{DE} S^{DE} = \frac{1}{2} (A_{DE} + A_{ED}) S^{DE} = 0 : \quad (5.8)$$

antisymmetric color words vanish, the former crossed word $W_{DE} - W_{ED}$ gives zero, and a nonzero result for the ordered pair (β, γ) must carry the symmetric color projection $f^{ACD} f^{BCE} S^{DE}$, which is indeed symmetric in $D \leftrightarrow E$.

The evanescent tensor integrals behind the loop coefficient are standard dimensional-reduction integrals [5]: with $d = 4 - 2\epsilon$, $\Delta > 0$, μ_ℓ^2 the evanescent part of the loop-momentum square, and $I_n(\Delta) := \int \frac{d^d \ell}{(2\pi)^d} \mu_\ell^2 (\ell^2 + \Delta)^{-n}$,

$$\lim_{\epsilon \rightarrow 0} I_3(\Delta) = \frac{1}{32\pi^2}, \quad \lim_{\epsilon \rightarrow 0} \Delta I_4(\Delta) = 0. \quad (5.9)$$

The tensor integral used in the former component-box evaluation is

$$J := \frac{1}{d} \int \frac{d^d \ell}{(2\pi)^d} \frac{\mu_\ell^2 \ell^2}{(\ell^2 + \Delta)^4} = \frac{1}{d} [I_3(\Delta) - \Delta I_4(\Delta)], \quad \lim_{\epsilon \rightarrow 0} J = \frac{1}{128\pi^2}, \quad \lim_{\epsilon \rightarrow 0} 2J = \frac{1}{64\pi^2}. \quad (5.10)$$

The first value is the tensor integral itself; the second includes the explicit factor 2 from the four-dimensional sigma/Fierz numerator. The two numbers cannot be identified with each other, and their separation rejects the former component-box evaluation and fixes its free sign and loop-counting power.

Why can the boxed formula (5.6) not be reproduced by an ordinary component graph? The locked component quadratic action pairs scalars with scalars (the $(\phi, \tilde{\phi})$ kinetic term) and fermions with fermions (the $(\psi, \tilde{\psi})$ kinetic term), so the mixed propagators vanish exactly,

$$G_{\phi_r \tilde{\psi}_s} = 0, \quad G_{\psi_r \tilde{\phi}_s} = 0. \quad (5.11)$$

A naive ordinary Wick box or triangle built from the two Yukawa families of (5.1), with external bottoms ψ_{r+} , $\tilde{\phi}_s$ and an output pair $\tilde{\lambda}\tilde{\lambda}$, would need precisely these scalar–fermion converting propagators; by (5.11) no such graph exists, and the ordinary two-Yukawa box is rejected as the parent of the (β, γ) anomaly. (An earlier argument had incorrectly promoted this ordinary-Wick obstruction to a statement about the regulated density divergence.)

What the nonzero (β, γ) anomaly actually is, at the component level, is the regulated coincident Euler Jacobian, differentiated before the bottom projection:

$$\tilde{\Phi}_s^B \frac{\vec{\delta}}{\delta \tilde{\Phi}_r^A} \longrightarrow \delta_{rs} K_{-, \epsilon}^{AB}(z, z) \longrightarrow \delta_{rs} f^{ACD} f^{BCE} \tilde{\mathcal{W}}_\alpha^D \tilde{\mathcal{W}}^{E\alpha}, \quad (5.12)$$

with the regulated coincident kernel

$$K_{-, \epsilon}^{AB}(z, z)|_{\tilde{\mathcal{W}}^2} = \frac{1}{32\pi^2} f^{ACD} f^{BCE} \tilde{\mathcal{W}}_\alpha^D \tilde{\mathcal{W}}^{E\alpha}. \quad (5.13)$$

Any future component realization must prove equality with (5.12); a nonzero ordinary box cannot be substituted for it. The supporting orbit evaluation fixes every factor: the regulated Schwinger–Dyson equation for the source vector field $X_{rs}^{AB} = \tilde{\Phi}_s^B \vec{\delta} / \delta \tilde{\Phi}_r^A$ shows that $r \neq s$ vanishes independently of any external target; the two opposite-sign superpotential occurrences have identical regulated contact kernels and cancel; the kinetic Euler occurrence is evaluated by (5.9) with no graph-specific factor $4 - d$; both color routings give the same tensor $f^{ACD} f^{BCE}$; and the interaction-copy factor is 1. Since $\partial_\alpha c = i\tilde{\mathcal{W}}_\alpha$, the dotted bilinear in (5.3) is minus the $\tilde{\mathcal{W}}$ bilinear in (5.13), which accounts for the overall sign of (5.6).

5.3 Open items

Remark 5.3 (Open and blocked items in the component route). Three component-route items remain outside the accepted result; none of them affects the boxed formula (5.6).

- (i) **The off-shell Noether identity is not accepted** (NOT_ACCEPTED_NOETHER_B1_B4_MISSING): the complete Euler operators and the announced termwise checks B1–B4 are absent; only the sign pattern of the proposed right-hand side is on record, unverified, so a component-level off-shell supersymmetry current with a fully verified divergence identity does not exist.
- (ii) **The Lorentzian vector cycle is blocked** (BLOCKED_STEP3D_LC_VECTOR_CYCLE): the Lorentzian vector integration cycle remains unfixed, so no Lorentzian component graph calculation can start.
- (iii) **The target-seeded component sweep of the 81 ordered letter pairs is not accepted** (NOT_ACCEPTED): the former sweep inherited the nonexistent ordinary two-Yukawa parent and classified pairs by comparison with a target-shaped table. The authoritative all-81 result is the target-blind superspace Schwinger-cut table of Section 4, with 29 nonzero and 52 zero ordered pairs.

The physical one-loop anomaly sector has no unresolved coefficient row; the open items above concern independent component re-derivations and the Lorentzian continuation only.

6 The heat-kernel approach

6.1 Motivation: a semigroup-based second derivation

The one-loop anomalous Ward identity of Section 4, with its universal coefficient $\frac{\hbar g^2}{16\pi^2}$, is derived perturbatively: a Schwinger–Dyson identity plus the cutting-failure mechanism of dimensional reduction, graph by graph. A second, independent and non-perturbative derivation of the same coefficient is highly desirable. The heat-kernel strategy would regulate the one-loop fluctuation determinant of the full $\mathcal{N} = 4$ Hessian complex by the semigroup e^{-sH} of a single elliptic operator H on the graded fluctuation bundle, and would extract the anomaly from a short-time Seeley–DeWitt coefficient of that semigroup [6]. Such a derivation would (i) cross-check the perturbative Schwinger-cut result without enumerating graphs; (ii) make the finiteness and locality of the anomaly manifest from short-time heat-kernel asymptotics; and (iii) connect directly to the evanescent structure of dimensional reduction through the same master integral $\frac{1}{32\pi^2}$ [5].

This section reports the exact status of that programme. The scalar auxiliary identities on which any such derivation must rest are established and machine-verified (Subsection 6.2). The full derivation, however, is blocked at the regulator level: no single elliptic generator on the full $\mathcal{N} = 4$ fluctuation complex has been constructed, and the anomaly coefficient used elsewhere in this paper is *not* obtained from this route. Subsection 6.3 states the obstruction precisely, and Subsection 6.4 lists what would close the route.

6.2 Established scalar and semigroup lemmas

Every identity in this subsection is exact; each is proved by direct scalar computation and is machine-verified in the source repository. We write $H(g) = H_0 + gH_1 + g^2H_2$ for the one-loop fluctuation operator expanded in the coupling, and $s > 0$ for the proper-time parameter.

Lemma 6.1 (Second-order Duhamel coefficient). *Let $E \rightarrow M$ be a fixed graded bundle, let $H_0 : E \rightarrow E$ be an even operator, and let H_1, H_2 be even endomorphism-valued differential operators. Set*

$$H(g) = H_0 + gH_1 + g^2H_2. \tag{6.1}$$

Then the coefficient of g^2 in the semigroup $e^{-sH(g)}$ is the two-term ordered integral

$$\begin{aligned} [g^2] e^{-sH(g)} = & - \int_0^s dt e^{-(s-t)H_0} H_2 e^{-tH_0} \\ & + \int_{0 < t_1 < t_2 < s} dt_1 dt_2 e^{-(s-t_2)H_0} H_1 e^{-(t_2-t_1)H_0} H_1 e^{-t_1H_0}. \end{aligned} \quad (6.2)$$

Remark 6.2 (The quadratic background block cannot be omitted). The first line of (6.2) is at the same background order as the second line. Hence a heat-kernel derivation of the physical coefficient must derive the full H_2 seagull, connection-square, gauge and ghost blocks; the quadratic background block cannot be omitted in favor of two linear H_1 insertions.

Lemma 6.3 (Ordered-simplex moment). *For integers $p, q \geq 0$,*

$$M_{p,q} := \int_{0 < a < b < 1} a^p b^q da db = \frac{1}{(p+1)(p+q+2)}. \quad (6.3)$$

Consequently, for $0 \leq k \leq m$ and $0 \leq \ell \leq n$, the tower coefficient is

$$T_{m,n;k,\ell} := \binom{m}{k} \binom{n}{\ell} M_{k+\ell, m+n-k-\ell} = \frac{\binom{m}{k} \binom{n}{\ell}}{(m+n+2)(k+\ell+1)}. \quad (6.4)$$

Proof. Integrating a at fixed b gives $M_{p,q} = \int_0^1 db b^q \frac{b^{p+1}}{p+1} = \frac{1}{(p+1)(p+q+2)}$, and (6.4) follows by substitution of (6.3). $\square$

Remark 6.4 (A second ordering supplies no automatic factor two). The moment (6.3) sums a single ordering of the simplex; no factor from the reversed ordering is included. A factor of two may be added only after an independent ordered-vertex calculation proves equality of the two orderings, including color order and Koszul sign.

Lemma 6.5 (Gaussian convolution and bridge means). *For $s > 0$ and $x \in \mathbb{R}^4$ let*

$$K_s(x) = \frac{1}{(4\pi s)^2} e^{-x^2/4s}. \quad (6.5)$$

Then K_s is normalized, solves the heat equation, and has diagonal value

$$\int_{\mathbb{R}^4} d^4x K_s(x) = 1, \quad \partial_s K_s = \Delta_{\mathbb{R}^4} K_s, \quad K_s(0) = \frac{1}{16\pi^2 s^2}, \quad (6.6)$$

and it obeys the convolution (Chapman–Kolmogorov) identity

$$\int_{\mathbb{R}^4} d^4y K_s(x-y) K_t(y) = K_{s+t}(x). \quad (6.7)$$

Moreover, for the three-segment bridge with $t_1, t_2, t_3 > 0$, $T = t_1 + t_2 + t_3$, and Gaussian weight $Q(x, y) = \frac{(x-w)^2}{4t_1} + \frac{(y-x)^2}{4t_2} + \frac{y^2}{4t_3}$ in one Cartesian component, the stationary means are

$$\langle x \rangle = \frac{t_2 + t_3}{T} w, \quad \langle y \rangle = \frac{t_3}{T} w, \quad (6.8)$$

and the inverse Hessian of Q is homogeneous of degree one under $(t_1, t_2, t_3) \mapsto (\rho t_1, \rho t_2, \rho t_3)$, $\rho > 0$.

Remark 6.6. Equation (6.8) proves the exact bridge mean and covariance scaling; it does not remove the endpoint factor $K_T(w)$.

Lemma 6.7 (Master-integral consistency with dimensional reduction). *Let $d = 4 - 2\varepsilon$, and let μ_ℓ^2 be the evanescent component of the loop momentum squared (Section 3). Then*

$$\lim_{\varepsilon \rightarrow 0} \mu^{2\varepsilon} \int \frac{d^d \ell}{(2\pi)^d} \frac{\mu_\ell^2}{(\ell^2 + \Delta)^3} = \frac{1}{32\pi^2}. \quad (6.9)$$

Remark 6.8. The value (6.9) agrees with the master integral of the accepted derivation (Sections 4 and 5), so the scalar heat-kernel layer is consistent with the evanescent mechanism of dimensional reduction [5]. Equality of this scalar integral does not, however, determine the $\mathcal{N} = 4$ operator pairing, the color tensor, the orientation sign, or the anomaly multiplicity.

Lemma 6.9 (Vanishing short-time limit at fixed nonzero shift). *For fixed $w \neq 0$,*

$$\lim_{s \downarrow 0} \frac{1}{(4\pi s)^2} e^{-w^2/4s} = 0. \quad (6.10)$$

Consequently a local derivative tower requires distributional coefficient extraction at $w = 0$; it cannot be obtained by deleting $K_s(w)$ before taking the limit.

6.3 The obstruction: the regulator proposal fails

The proposed continuation defines its would-be generator blockwise from a Hessian two-form $\delta^2 S$ and a field-space identification $\mathcal{G}$. Its own displayed free matter blocks give

$$(\mathcal{G}^{-1} \delta^2 S)^2 \Big|_{E_+} = \bar{\nabla}^2 \nabla^2 = 16 \square \mathcal{P}_+, \quad (6.11)$$

whereas the declared generator is $-\square \mathcal{P}_+$; the accompanying verifier inserts a factor $-1/16$ that appears nowhere in the proposal. Both the normalization (16) and the sign are wrong relative to the declared generator. Likewise, the proposal's own vector-block weights give

$$\mathcal{G}_V^{-1} \delta^2 S_{VV} = (2g^2) \frac{1}{2g^2} \square = \square, \quad (6.12)$$

whereas the declared generator is $-\square$; the verifier adds an undeclared Euclidean sign. Finally, the prescription applies the squared operator on the matter and ghost rows but the unsquared operator on the vector row. This is not a single operator functional calculus when nonzero vector-matter mixing blocks are retained: different powers on different blocks define different operators, so the off-diagonal mixing blocks have no defined heat evolution under this prescription.

Remark 6.10 (Status of the route: blocked). For the three independent reasons above, the full heat-kernel derivation of the anomaly coefficient is currently blocked; the source repository records the status `BLOCKED_HEAT_KERNEL_TYPED_REGULATOR_AND_COEFFICIENT_DERIVATION`. The identities of Subsection 6.2 are admitted as verified auxiliary lemmas only: they do not define a regulator on the full $\mathcal{N} = 4$ fluctuation complex, and they do not independently derive the coefficient $\frac{hg^2}{16\pi^2}$, which remains the result of the Schwinger-cut derivation of Section 4.

6.4 What would close the route

The route would produce a regulator only when the following are derived together, not declared:

- a *single elliptic generator* on the full graded $\mathcal{N} = 4$ fluctuation bundle—one operator, one functional calculus, with the correct sign and normalization on every block (matter, vector, ghost), covering also the off-diagonal vector-matter mixing blocks;

- its *dual Euler action* on the letter bilinears, consistent with the Schwinger–Dyson/Euler reduction of Section 4;
- the *full quadratic coefficient* H_2 —the seagull, connection-square, gauge and ghost blocks at order g^2 —which by Lemma 6.1 sits at the same background order as two linear insertions and cannot be dropped;
- the *distributional short-time limit*: coefficient extraction at $w = 0$ in the distributional sense, as required by Lemma 6.9.

Reproducing the accepted coefficient $\frac{\hbar g^2}{16\pi^2}$ with the correct color tensor and the correct census of nonzero channels would in addition require the $\mathcal{N} = 4$ operator pairing, the orientation sign and the anomaly multiplicity to be derived from that generator, since none of them follow from the verified scalar identities alone. Until then, the heat-kernel approach stands as an auxiliary consistency layer, not as a second derivation.

7 Comparison with the holomorphic twist

The one-loop coefficients of Section 4 were derived, validated, and sealed before the holomorphic-twist literature for this problem was consulted. The present section compares that sealed result with the independent prediction of Budzik and Kulp [1], who compute the one-loop correction to the supersymmetry algebra of the holomorphic twist of $\mathcal{N} = 4$ super-Yang–Mills theory. The comparison is an external cross-check: no coefficient of [1] enters the derivation of Section 4, and everything concluded here is collected in the admissibility statement of Subsection 7.5.

7.1 The external target

In the holomorphic twist [1, 7] the $\mathcal{N} = 1$ vector multiplet becomes a holomorphic bc system and each chiral multiplet a holomorphic $\beta\gamma$ system on $\mathbb{C}^2$. The Batalin–Vilkovisky fields are

$$\begin{aligned} b &= b^{(0)} + b^{(1)} + b^{(2)} \in \Omega^{0,\bullet}(\mathbb{C}^2, \mathfrak{g}^\vee), & c &= c^{(0)} + c^{(1)} + c^{(2)} \in \Omega^{0,\bullet}(\mathbb{C}^2, \mathfrak{g})[1], \\ \beta_I &= \beta_I^{(0)} + \beta_I^{(1)} + \beta_I^{(2)} \in \Omega^{0,\bullet}(\mathbb{C}^2, R^\vee)[1], & \gamma^I &= \gamma^{I(0)} + \gamma^{I(1)} + \gamma^{I(2)} \in \Omega^{0,\bullet}(\mathbb{C}^2, R), \end{aligned} \quad (7.1)$$

with action (their eq. (3.1))

$$S_{\text{BV}} = \int_{\mathbb{C}^2} d^2z \left\{ \text{Tr} \left[b \left(\bar{\partial}c + \frac{1}{2}[c, c] \right) \right] + \beta_I (\bar{\partial}\gamma^I + c\gamma^I) + W[\gamma] \right\}. \quad (7.2)$$

The tree differential acts as

$$\begin{aligned} Q_0 c &= \frac{1}{2}[c, c], & Q_0 b &= [c, b] - \mu(\beta, \gamma), \\ Q_0 \gamma^I &= c\gamma^I, & Q_0 \beta_I &= \beta_I c + \partial_{\gamma^I} W, \end{aligned} \quad (7.3)$$

with $\mu(\beta, \gamma)$ the moment map. The letter gradings are $|b| = |\gamma| = 0$ and $|c| = |\beta| = 1 \bmod 2$, and normal ordering is supercommutative on letters. The untwisted identifications are $\gamma \sim \bar{\phi}$, $\beta \sim \psi_+$, $\partial_{\dot{\alpha}} c \sim \bar{\lambda}_{\dot{\alpha}}$, $b \sim F_{++}$, and $\partial_{\dot{\alpha}} \sim D_{+\dot{\alpha}}$, consistent with the component table of Section 5. The Lie algebra is normalized as (their eq. (3.11))

$$[t_A, t_B] = f_{AB}^C t_C, \quad \text{Tr}(t_A t_B) = \kappa_H \delta_{AB}, \quad \text{Tr}(t_A [t_B, t_C]) = \kappa_H f_{ABC}, \quad (7.4)$$

with κ_H the dual Coxeter number; the missing factor of i relative to our convention $[T_A, T_B] = i f_{AB}^C T_C$ is supplied by the frame map (7.11) below. For the $\mathcal{N} = 4$ theory the matter representation is $R = \mathfrak{g} \otimes \mathbb{C}^3$, the superpotential is $W[\gamma] = \text{Tr} \gamma^1 [\gamma^2, \gamma^3] = \frac{1}{3!} \varepsilon_{IJK} \kappa_H f_{ABC} \gamma^{IA} \gamma^{JB} \gamma^{KC}$,

the twisting supercharge is Q_-^4 , and $\gamma^I \sim \Phi^{4I}$, $\beta_I \sim \Psi_{I+}$, $\partial_{\dot{\alpha}} c \sim \bar{\Psi}_{\dot{\alpha}}^4$, $b \sim F_{++}$. The 1/16-BPS fields assemble into a single superfield on $\mathbb{C}^{2|3}$ (their eq. (3.56)),

$$C = c + \theta_I \gamma^I + \frac{1}{2} \varepsilon^{IJK} \theta_I \theta_J \beta_K + \theta_1 \theta_2 \theta_3 b, \quad (7.5)$$

in terms of which the action is a holomorphic Chern–Simons theory,

$$S_{\text{BV}} = \frac{1}{2} \int_{\mathbb{C}^{2|3}} d^2 z d^3 \theta \operatorname{Tr} \left[C \left(\bar{\partial} C + \frac{1}{3} [C, C] \right) \right]. \quad (7.6)$$

The classical supercharge obeys the tree identity (their eq. (1.9))

$$Q_0 C(\theta) = \frac{1}{2} [C(\theta), C(\theta)]. \quad (7.7)$$

Their one-loop result is a compact prediction for the loop-corrected supercharge Q_1 (their eqs. (1.10) and (3.69)),

$$Q_1(C^A(\theta) C^B(\theta')) = -\kappa_H^2 f_{ACD} f_{BCE} (\theta_1 - \theta'_1)(\theta_2 - \theta'_2)(\theta_3 - \theta'_3) \partial_{\dot{\alpha}} C^D(\theta) \partial^{\dot{\alpha}} C^E(\theta'). \quad (7.8)$$

(The introduction of [1] prints the coefficient $-1/4$ where its main text prints $-\kappa_H^2$; this is the second source inconsistency, see Subsection 7.4.) Expanded in the components of (7.5), the prediction reads (their eqs. (3.63)–(3.68))

$$\begin{aligned} Q_1(b^A c^B) &= \kappa_H^2 f_{ACD} f_{BCE} \partial_{\dot{\alpha}} c^D \partial^{\dot{\alpha}} c^E, \\ Q_1(b^A b^B) &= \kappa_H^2 f_{ACD} f_{BCE} [\partial_{\dot{\alpha}} c^D \partial^{\dot{\alpha}} b^E - \partial_{\dot{\alpha}} b^D \partial^{\dot{\alpha}} c^E + \partial_{\dot{\alpha}} \beta_I^D \partial^{\dot{\alpha}} \gamma^{IE} - \partial_{\dot{\alpha}} \gamma^{ID} \partial^{\dot{\alpha}} \beta_I^E], \\ Q_1(\beta_I^A \gamma^{JB}) &= \kappa_H^2 \delta_I^J f_{ACD} f_{BCE} \partial_{\dot{\alpha}} c^D \partial^{\dot{\alpha}} c^E, \\ Q_1(\beta_I^A \beta_J^B) &= \kappa_H^2 \varepsilon_{IJK} f_{ACD} f_{BCE} [\partial_{\dot{\alpha}} c^D \partial^{\dot{\alpha}} \gamma^{KE} - \partial_{\dot{\alpha}} \gamma^{KD} \partial^{\dot{\alpha}} c^E], \\ Q_1(b^A \gamma^{IB}) &= \kappa_H^2 f_{ACD} f_{BCE} [\partial_{\dot{\alpha}} c^D \partial^{\dot{\alpha}} \gamma^{IE} - \partial_{\dot{\alpha}} \gamma^{ID} \partial^{\dot{\alpha}} c^E], \\ Q_1(\beta_I^A b^B) &= \kappa_H^2 f_{ACD} f_{BCE} [\partial_{\dot{\alpha}} \beta_I^D \partial^{\dot{\alpha}} c^E + \partial_{\dot{\alpha}} c^D \partial^{\dot{\alpha}} \beta_I^E + \varepsilon_{IJK} \partial_{\dot{\alpha}} \gamma^{JD} \partial^{\dot{\alpha}} \gamma^{KE}], \end{aligned} \quad (7.9)$$

while $Q_1(cc) = Q_1(\gamma\gamma) = Q_1(\gamma c) = Q_1(\beta c) = 0$. In the L_∞ language of [1] the $(\ell + 2)$ -ary bracket is the ℓ -loop correction, with $\{\mathcal{I}, \cdot\}_0 = Q_0$ and $\frac{1}{2!} \{\mathcal{I}, \mathcal{I}, \cdot\}_0 = Q_1$; the Maurer–Cartan identities give $\{Q_0, Q_1\} = 0$ and $Q_1^2 + \{Q_0, Q_2\} = 0$. Hence Q_1 is nilpotent only on Q_0 -cohomology and higher corrections Q_n are necessarily present: the construction of [1] is not one-loop exact, and the comparison below concerns its one-loop operation Q_1 .

7.2 The dictionary

Let R_A^P be a color frame, U_I^r a flavor frame, $S_{\dot{a}}^{\dot{b}}$ a dotted-spin frame, and $\rho \neq 0$ an overall scale. The exact map between the letters $b, \beta_r, \gamma^r, \partial_{\dot{\alpha}} c$ of Section 4 and the alphabet of [1] is

$$\begin{aligned} b^A &\mapsto \rho R_A^P b^P, \\ \beta_I^A &\mapsto \rho R_A^P U_I^r \beta_r^P, \\ \gamma^{IA} &\mapsto \rho R_A^P (U^{-1})^I_r \gamma^{rP}, \\ \partial_{\dot{\alpha}} c^A &\mapsto \rho R_A^P S_{\dot{a}}^{\dot{b}} \partial_{\dot{b}} c^P, \end{aligned} \quad (7.10)$$

where the left-hand sides are the fields of (7.1) and the right-hand sides use the letters of Section 4. Every row carries the common scale ρ : the factors $-i/\sqrt{2}$ and $1/\sqrt{2}$ familiar from the raw superfield composites are precisely those absorbed into the letter definitions of Definition 4.1, so that in the letters of this paper the map is uniform. The γ row carries the inverse flavor frame and the ∂c row the spin frame. At $\rho = 1$ with identity frames the map reduces to

the identity, $b \mapsto b$, $\beta_I \mapsto \beta_r \delta_I^r$, $\gamma^I \mapsto \gamma^r \delta^I_r$, and $\partial_{\dot{a}} c \mapsto \partial_{\dot{a}} c$. The frames enter the color and trace relations through

$$R_A^P R_B^Q \delta_{PQ} = \delta_{AB}, \quad t_A = -i R_A^P T_P, \quad f_{AB}^C = R_A^P R_B^Q f_{PQ}^R (R^{-1})_R^C, \quad \text{Tr}_H = -\kappa_H \text{Tr}, \quad (7.11)$$

where Tr_H is the trace of (7.4) and Tr ours, with $\kappa_{AB} = \delta_{AB}$. Applied to (7.5), the map sends

$$C(\theta) \mapsto \rho \left[c + \theta_r \gamma^r + \frac{1}{2} \varepsilon_{rst} \theta_r \theta_s \beta_t + \theta_1 \theta_2 \theta_3 b \right]; \quad (7.12)$$

the first entry is the ghost (the formal ghost placeholder of the letter algebra), which takes part in the twisted theory but not in the physical letter pairs compared below. Matching the one-loop coefficients of the two formalisms on the gaugino ($\mathcal{W}_+$) component gives the loop-scale relation

$$\hbar_{\text{HT}} \kappa_H^2 = \frac{\hbar g^2}{32\sqrt{2}\pi^2} = \frac{1}{2\sqrt{2}} \frac{\hbar g^2}{16\pi^2}. \quad (7.13)$$

Remark 7.1 (conditional status of (7.13)). The relation (7.13) is obtained under a conditional total-differential scale in the dictionary analysis; it is a conditional dictionary entry, not a theorem proved in this paper. The four physical letter rows of (7.10) are independent of this conditionality, and the comparison of Subsection 7.3 is performed on output words only. A single κ_H presupposes a simple gauge algebra; for a general reductive gauge algebra the trace relations hold factorwise.

7.3 Comparison results

The comparison proceeds word by word. Each ordered pair of the nine letters $(b, \beta_r, \gamma^r, \partial_{\dot{a}} c)$ is mapped by (7.10); the predicted output monomials are translated into the letters of Section 4; and the result is compared with the sealed coefficient table over $\mathbb{Q}(i, \sqrt{2})$. With the identification $Q_0 \leftrightarrow -\frac{1}{2}\nabla_-$ of Section 4, the quantity compared with $Q_1(XY)$ is the one-loop anomalous variation $\nabla_-(XY)|_{1\text{-loop}}$, normalized by (7.13).

Proposition 7.2 (ordered-word comparison). *For every ordered pair of letters, the sealed coefficient word of Section 4, its independent expansion, and the translated holomorphic-twist word agree exactly:*

$$N_{\text{pairs}} = 81, \quad N_{\text{agree}} = 81, \quad N_{\text{disagree}} = 0, \quad 81 = 29_{\text{nonzero}} + 52_{\text{zero}}. \quad (7.14)$$

The nonzero census decomposes as

$$29 = \frac{1}{(b,b)} + \frac{6}{(b,\beta_r),(\beta_r,b)} + \frac{6}{(b,\gamma^r),(\gamma^r,b)} + \frac{6}{(\beta_r,\beta_s),r \neq s} + \frac{6}{(\beta_r,\gamma^r),(\gamma^r,\beta_r)} + \frac{4}{(b,\partial_{\dot{a}}c),(\partial_{\dot{a}}c,b)}, \quad (7.15)$$

comprising $8 + 6 \cdot 4 + 6 \cdot 2 + 6 \cdot 2 + 6 \cdot 1 + 4 \cdot 2 = 70$ nonzero output words; the remaining 52 ordered pairs vanish on both sides.

The mixed rows $(b, \partial_{\dot{a}} c)$ and $(\partial_{\dot{a}} c, b)$ display the intrinsic split $K_{(e_{\dot{a}});0} = \frac{2}{3}$ and $K_{(e_{\dot{a}});(e_{\dot{a}})} = \frac{1}{3}$ for a unit dotted derivative $e_{\dot{a}}$, the two orderings being exchanged by the Leibniz rule; this is the $\frac{1}{3} - \frac{2}{3}$ distribution already recorded in the channel table of Section 4. At arbitrary derivative order the comparison uses the Appendix-B tower of [1] (their eq. (B.5)),

$$m! n! C_{mn} = \frac{1}{m+n+2} \sum_{k=0}^m \sum_{\ell=0}^n \frac{1}{k+\ell+1} \binom{m}{k} \binom{n}{\ell} (p_1)_1^k (p_1)_2^\ell (p_2)_1^{m-k} (p_2)_2^{n-\ell}, \quad (7.16)$$

whose component formulas (their eqs. (B.6)–(B.11)) multiply this tower by $\kappa_H^2 f_{ACD} f_{BCE}$ with the same flavor structures as the zero-derivative pairs (7.9); we write their holomorphic form variables as p_i , $p_1 + p_2 + p_3 = 0$, to avoid a clash of notation.

Proposition 7.3 (all-orders kernel identity). *For all $m, n \in \mathbb{Z}_{\geq 0}$ and $0 \leq k \leq m$, $0 \leq \ell \leq n$,*

$$K_{m,n;k,\ell} = \frac{2 \binom{m}{k} \binom{n}{\ell}}{(m+n+2)(k+\ell+1)} = 2 T_{m,n;k,\ell}^{\text{HT,printed}} = T_{m,n;k,\ell}^{\text{HT,corrected}}, \quad (7.17)$$

where $K_{m,n;k,\ell}$ is the derivative kernel of the one-loop variation of Section 4,

$$T_{m,n;k,\ell}^{\text{HT,printed}} = \frac{\binom{m}{k} \binom{n}{\ell}}{(m+n+2)(k+\ell+1)} \quad (7.18)$$

is the coefficient tower of (7.16), and $T^{\text{HT,corrected}} := 2 T^{\text{HT,printed}}$. At the origin $T_{0,0;0,0}^{\text{HT,printed}} = \frac{1}{2}$ while $K_{0,0;0,0} = 1$.

The factor 2 in (7.17) is the $\Gamma(3) = 2$ of the triangle Feynman parametrization,

$$\frac{1}{D_0 D_1 D_2} = 2 \int_{\Delta_2} \frac{1}{(r^2 + \Delta)^3}, \quad 2 \int_{\Delta_2} 1 = 1, \quad (7.19)$$

not an orientation multiplicity. The triangle operator of [1] (their eq. (3.8)), with $\partial = (\partial_1, \partial_2)$ the holomorphic derivative on $\mathbb{C}^2$,

$$\mathcal{D}_{w,z}^\Delta(f, g) := \sum_{m,n=0}^{\infty} \frac{1}{(n+m+2)!} e^{z \cdot \partial} ((w-z) \cdot \partial)^n \partial_{\dot{\alpha}} f(y) e^{w \cdot \partial} ((z-w) \cdot \partial)^m \partial^{\dot{\alpha}} g(y) \Big|_{y=0}, \quad (7.20)$$

generates the generic shifted pairs, each Q_1 pair action being $\kappa_H^2 f_{ACD} f_{BCE}$ times the corresponding $\mathcal{D}^\Delta$ bilinear; its printed zero-shift value (their eq. (3.9)),

$$\mathcal{D}_{0,0}^\Delta(f, g) = \frac{1}{2} \partial_{\dot{\alpha}} f \partial^{\dot{\alpha}} g, \quad (7.21)$$

misses precisely the prefactor 2 of (7.19), while the printed component formulas (7.9) carry the complete-amplitude normalization (see Subsection 7.4). The identity (7.17) is proved symbolically for all $m, n \geq 0$ and checked coefficient by coefficient on the finite rectangle $0 \leq m, n \leq 8$:

$$\sum_{m=0}^8 \sum_{n=0}^8 (m+1)(n+1) = \left(\sum_{j=1}^9 j \right)^2 = 45^2 = 2025 \quad \text{kernel checks, all passing (2025/2025)}. \quad (7.22)$$

Multiplication by the 70 nonzero base output words lifts this to the full derivative tower,

$$2025 \times 70 = 141750 \quad \text{lifted coefficient checks, all passing (141750/141750)}, \quad (7.23)$$

while the 52 vanishing rows remain zero at every derivative order. The independent physical verification suite accepts the physical one-loop anomaly sector with 33/33 checks passed and 0 failed.

7.4 Two source-internal inconsistencies and their adjudication

The source [1] contains two internal normalization conflicts. Both are quarantined as matters of source provenance: the coefficients of this paper are derived independently in Section 4, and the comparison above uses only output words, so no contested normalization of [1] is imported.

The factor two. Printed inside [1] are: the zero-shift kernel (7.21), carrying $1/2 = 1/\Gamma(3)$; the shifted-pair formulas multiplying $\mathcal{D}^\Delta$ by κ_H^2 , hence $\kappa_H^2/2$ at $z = w = 0$; the zero-shift component formulas (7.9), printing κ_H^2 times the derivative bilinears; and the tower (7.16), printing $\kappa_H^2/(m + n + 2) = \kappa_H^2/2$ at $m = n = 0$. The evaluated branches give

$$\mathcal{I}_\Delta[p; 0] = \mathcal{D}_{0,0}^\Delta = C_{00} = \frac{1}{2} , \quad \frac{c_{\text{component}}}{c_{\text{triangle}}} = \frac{c_{\text{compact}}}{c_{\text{triangle}}} = 2 , \quad (7.24)$$

with $\mathcal{I}_\Delta[p; 0] = \frac{1}{2} p_1 \wedge p_2$ the zero-shift master triangle (their eqs. (2.35) and (2.39)). The adjudication selects the complete-amplitude branch: the zero-shift component block and the compact block of [1] carry the complete normalization, while its master-integral, generic shifted-pair, and Appendix-B branches are uniformly smaller by the factor 2 and must be multiplied by 2 before comparison with the complete amplitude, consistently with (7.17) holding at every derivative degree. The conflict remains unresolved inside the source: the classification recorded here repairs nothing in [1] and changes no relative coefficient of Section 4.

The compact coefficient. The introduction of [1] prints the compact prediction (7.8) with coefficient $-1/4$; its main text prints $-\kappa_H^2$. The two displays agree only if

$$\kappa_H^2 = \frac{1}{4} . \quad (7.25)$$

The independently fixed coefficient of Section 4 selects the main-text branch $-\kappa_H^2$ and rejects the introductory display unless $\kappa_H^2 = 1/4$; again this classifies the conflict without repairing the source.

7.5 The admissibility verdict

Theorem 7.4 (admissibility of the holomorphic-twist comparison). *Under the dictionary (7.10), the identification $Q_0 \leftrightarrow -\frac{1}{2}\nabla_-$ of Section 4, and the loop-scale relation (7.13), the direct superspace calculation of Section 4 exactly reproduces the holomorphic-twist prediction for the one-loop semi-chiral anomaly: the 81 ordered output words agree as in (7.14) (29 nonzero, 52 zero), and the derivative kernels agree to all orders,*

$$K_{m,n;k,\ell} = T_{m,n;k,\ell}^{\text{HT,corrected}} \quad \text{for all } m, n \geq 0 , \quad (7.26)$$

i.e. the sealed one-loop coefficient table coincides with the factor-two-corrected one-loop supercharge Q_1 of [1] on every output word, with no mismatch.

Remark 7.5 (agreement of totals is not an absence proof). Theorem 7.4 compares total coefficients. Writing $C_{\text{complete}} = C_{\text{tabulated}} + \sum_{j \in \mathcal{O}} c_j$ for the contributions c_j of any set $\mathcal{O}$ of omitted Wick routings,

$$C_{\text{tabulated}} = C_{\text{HT}} \quad \text{does not imply} \quad \mathcal{O} = \emptyset \text{ or } c_j = 0 \text{ for every } j : \quad (7.27)$$

cancellations such as $c_1 = t, c_2 = -t$ with $t \neq 0$ leave the total unchanged. Equality of the total coefficient with the holomorphic-twist prediction therefore does not prove termwise absence of omitted Wick routings; the agreement is a final cross-check, not a target-blind absence proof.

Remark 7.6 (the adversarial census remains open). The completeness question for the graph census is not settled by Theorem 7.4 and carries the status `OPEN_FINAL_CENSUS_17_CANDIDATES`: the ghost–Nielsen–Kallosh sector has not run, the zero/cut-orbit sector has not run, the automated three-lens refutation has not run, and four physics-relevant Wick routings remain deferred to a future per-routing derivation from the locked action, without reading the target.

Remark 7.7 (the accepted coefficients are unchanged). The accepted coefficient table of Section 4 rests on the target-blind Schwinger-cut derivation presented there; it was read, validated, and sealed (SHA-256 `ec77327d...b3016d`) before the source of [1] was consulted, and it is unaffected by the open census: within the physical one-loop anomaly sector no coefficient row is unresolved.

8 Conclusions and open problems

We have determined the complete one-loop anomalous supersymmetry transformation law of the BPS-letter composites in Euclidean $\mathcal{N} = 4$ super-Yang–Mills theory quantized in $\mathcal{N} = 1$ superspace. For each of the 81 ordered letter pairs, the semi-chiral derivative of the composite equals the universal coefficient $\hbar g^2/(16\pi^2)$ times the ordered color tensor $f^{ACD}f^{BCE}$ times a fixed, explicitly known letter bilinear: 29 channels are nonzero and 52 vanish exactly, with the zeros accounted for by four independent structural mechanisms (pairs with no tree descendant, flavor-off-diagonal mixed pairs, repeated-index diagonal pairs, and pairs killed by the charge and vertex analysis). The sole microscopic origin of the anomaly is the evanescent failure of the DRED cutting rule, (1.1): one universal μ_ℓ^2 insertion per marked line, whose master integral is the finite constant $1/(32\pi^2)$. The anomaly is therefore ultraviolet-finite, local in the external momenta, and — after a finite normal-product term fixed by the residual supersymmetries, under which the nonzero channels close — scheme-independent. Resummed over the holomorphic derivatives, the anomaly is governed by the closed-form kernel (1.3), whose prefactor is the $\Gamma(3) = 2$ of the triangle Feynman parametrization.

Three independent bodies of evidence support the result. First, the superspace derivation is target-blind: every channel was computed from the locked action by the Schwinger–Dyson reduction and the cutting failure, with exact arithmetic, before any external target was consulted. Second, the holomorphic-twist comparison of Sec. 7 is exact: all 81 ordered output words and all checked tower coefficients agree with the one-loop prediction of [1] once its printed tower coefficient is corrected by the factor 2, and the same comparison selects the main-text branch of the source’s compact coefficient. Third, the component bottom projection (1.4) provides an exact component-level corollary, with the color-symmetric dotted-gaugino bilinear. The heat-kernel route contributes exact auxiliary lemmas but not a second derivation.

Remark 8.1. The scope of the accepted result is the physical one-loop anomaly sector in Euclidean signature. The holomorphic-twist agreement is an external cross-check, not an internal completeness proof, and no Lorentzian-signature statement is made.

Open problems

- **Adversarial census completeness.** Seventeen candidate graph sectors or routings raised by the adversarial census review remain adjudicated: the ghost–Nielsen–Kallosh sector, the zero/cut-orbit sector, the automated refutation pass, and four physics-relevant Wick routings have not been derived from the locked action without reference to the target. Status: `OPEN_FINAL_CENSUS_17_CANDIDATES`. This does not alter any accepted coefficient; it is the remaining completeness-type question, since equality with the external target cannot by itself exclude omitted contributions that cancel in the total.
- **Heat-kernel second derivation.** A non-perturbative derivation requires, derived together rather than declared: a single elliptic generator with the correct sign and normalization on every block (matter, vector, ghost, and the off-diagonal mixing blocks), a field-space super-metric, the full second-order coefficient of the semigroup expansion, and distributional short-time extraction at coincident points. Status: `BLOCKED_HEAT_KERNEL_TYPED_REGULATOR_AND_COEFFICIENT_I`.
- **Component off-shell Noether identity.** The component-level off-shell supersymmetry current whose divergence reproduces the anomalous transformation law term by term has not been established: the complete component Euler operators and the announced termwise checks are absent. Status: `NOT_ACCEPTED_NOETHER_B1_B4_MISSING`. The complete component Feynman rules and an independent all-pairs component sweep are likewise not accepted; the only accepted component statement is the exact bottom projection (1.4).

- **Lorentzian vector integration cycle.** The Lorentzian vector integration cycle of the regulated path-integral measure remains unfixed. Status: `BLOCKED_STEP3D_LC_VECTOR_CYCLE`.
- **Lorentzian sector.** All results of this paper are Euclidean. The Lorentzian component and Lorentzian superspace calculations are gated on a frozen Euclidean bridge and on the vector cycle above; the continuation of the anomalous Ward identity to Lorentzian signature is deferred.

Remark 8.2. The holomorphic-twist algebra of [1] requires higher-loop corrections (the one-loop operator squares to zero only on the tree cohomology). Extending the present exact one-loop census of the anomalous supersymmetry transformation law beyond one loop is an open problem on both sides of the comparison.

A Verification status

All coefficient-level statements of this paper are backed by machine-locked exact-arithmetic checks, performed over the rational field extended by i and $\sqrt{2}$ (no floating-point approximation at any stage). This appendix records what was checked, what the independent reviews concluded, and what remains open. Tokens in typewriter font are verbatim status strings of the underlying audits.

Machine-locked exact-arithmetic checks

- **81-channel census.** The target-blind 81-row coefficient table of Sec. 4: all 81 ordered channels complete and exact, 29 nonzero and 52 exactly zero, 0 open and 0 unresolved rows; the table was read, validated, canonicalized and sealed (SHA-256) before the external holomorphic-twist target was read.
- **Seed-channel replays.** The field-strength seed channel: 269 external-slot replay checks, 0 failures. Color-mask replay: 9216 full color-mask checks plus a 2048-entry sparse replay, 0 equality failures.
- **DRED audits.** The cutting-failure identity (1.1) for every marked line, and the evanescent μ_ℓ^2 -moment master integrals — the triangle value $1/(32\pi^2)$ and the rank-two values $J = 1/(128\pi^2)$, $2J = 1/(64\pi^2)$ — are locked in a machine-regenerated master-integral table, including the identities $\Delta J_3 = (\epsilon/2)J_2$ and $\text{Res}_{\epsilon=0} J_2 = 1/(16\pi^2)$.
- **Sector checks.** Finite residual-Ward checks of the mixed-chirality channels: 31/31. Regulated Schwinger–Dyson audit of the channel underlying (1.4): 84/84. Epsilon-canonicalization audit of the mixed flavor sector: 42/42. Heat-kernel auxiliary lemmas (Gaussian semigroup, bridge moments, short-time limit, simplex moment, Duhamel coefficient, DRED master integral): 8/8.
- **Holomorphic-twist round trip.** Direct ordered output-word matches: 81/81, 0 mismatches. Finite-rectangle kernel checks on $0 \leq m, n \leq 8$: 2025/2025 ($= 45^2$), 0 mismatches. Lifted to all 70 nonzero base output words: $2025 \times 70 = 141750/141750$ lifted coefficient checks, 0 mismatches. The external source import itself passed 267 consistency checks.
- **Physical verifier.** 33/33 checks passed, 0 failed; status `ACCEPTED`, accepted scope `PHYSICAL_ONE_LOOP_ANOMALIES`.

Independent review trail

An independent physics re-derivation review of the one-loop result was conducted on 2026-07-16 (non-authority review evidence). Its spot-check verdict, `SPOT_CHECKS_PASSED_WITH_BOUNDED_RESIDUAL_SURFA` confirms the seed-channel derivation chain, the 81-channel census, and the factor-two adjudication, leaving four named bounded residual surface items, none of which is a defect in any accepted coefficient. An external review (GPT Pro, Gate-15) independently returned `FINAL_ONE_LOOP_HT_SETTLEMENT_ACCEPTED` for the physical one-loop anomaly sector, covering the universal cutting-failure identity, the exact census $29 = 1 + 6 + 6 + 6 + 6 + 4$ with $81 - 29 = 52$, the residual-Ward-compatible finite renormalization of the mixed-chirality channels, the anchor-free scale fixing, and the holomorphic-twist kernel identity.

Remark A.1. The adversarial-census review remains open: `OPEN_FINAL_CENSUS_17_CANDIDATES`. Its proposed completeness certificate — exact agreement with the holomorphic-twist target — was rejected, because equality of totals, $C_{\text{enum}} = C_{\text{HT}}$, does not imply an empty omission set: additional routings with contributions c_j satisfying $\sum_j c_j = 0$ would leave the total unchanged, e.g. $c_1 = t, c_2 = -t$ with $t \neq 0$. The holomorphic-twist agreement is therefore a final cross-check, not a target-blind absence proof. The accepted coefficient table is unaffected: its acceptance rests on the independent target-blind superspace derivation, not on either census proposal.

Remark A.2. The machine checks verify exact-arithmetic equalities, replays of locked derivation chains, and target comparisons; they are not by themselves an independent re-derivation. The independent reviews cited above are advisory evidence; the authority for the accepted result is the derivation of Sec. 4 itself.

References

- [1] K. Budzik and J. Kulp, *Loop Corrected Supercharges from Holomorphic Anomalies*, arXiv:2512.07771.
- [2] S. Weinberg, *The Quantum Theory of Fields, Vol. III: Supersymmetry*, Cambridge University Press, 2000, Chapter 30.
- [3] S. J. Gates, M. T. Grisaru, M. Roček and W. Siegel, *Superspace, or One Thousand and One Lessons in Supersymmetry*, Front. Phys. **58** (1983) 1, arXiv:hep-th/0108200.
- [4] I. A. Batalin and G. A. Vilkovisky, *Gauge Algebra and Quantization*, Phys. Lett. B **102** (1981) 27.
- [5] W. Siegel, *Supersymmetric Dimensional Regularization via Dimensional Reduction*, Phys. Lett. B **84** (1979) 193.
- [6] D. V. Vassilevich, *Heat kernel expansion: user's manual*, Phys. Rept. **388** (2003) 279, arXiv:hep-th/0306138.
- [7] K. Costello, *Notes on supersymmetric and holomorphic field theories in dimensions 2 and 4*, arXiv:1111.4234.